

Contents

1 Basic	1	6 Math	12
1.1 Default Code	1	6.1 Characteristic Polynomial	12
1.2 Debugger	1	6.2 Discrete Logarithm	12
1.3 Fast IO	1	6.3 Extgcd	12
1.4 vimrc	1	6.4 Floor Sum	12
2 Graph	1	6.5 Factorial Mod P^k	12
2.1 SCC	1	6.6 Gaussian Elimination	13
2.2 BCC	2	6.7 Linear Function Mod Min	13
2.3 Centroid Decomposition	2	6.8 MillerRabin PollardRho	13
2.4 Count Cycles	2	6.9 Lucy Hedgehog	13
2.5 DirectedMST	2	6.10 Quadratic Residue	14
2.6 Dominator Tree	3	6.11 Simplex	14
2.7 Heavy Light Decomposition	3	6.12 Linear Programming Construction	14
2.8 Virtual Tree	3	6.13 Estimation	14
2.9 Vizing	3	6.14 Theorem	15
2.10 Maximum Clique Dynamic	4	6.15 General Purpose Numbers	15
2.11 Minimum Steiner Tree	4	6.16 Integral	15
2.12 Theory	5	6.17 Floor Sum	15
3 Data Structure	5	7 Polynomial	16
3.1 Dynamic Line Hull	5	7.1 FFT	16
3.2 Leftist Tree	5	7.2 NTT	16
3.3 Link Cut Tree	5	7.3 FWT	16
3.4 Splay Tree	6	7.4 Polynomial	16
4 Flow/Matching	6	7.5 BM-BM	18
4.1 Hopcroft Karp	6	7.6 Generating Functions	19
4.2 Dinic	6	8 Geometry	19
4.3 Min Cost Max Flow	7	8.1 Basic	19
4.4 Min Cost Circulation	7	8.2 Convex Hull	19
4.5 Kuhn Munkres	8	8.3 Dynamic Convex Hull	19
4.6 Stoer Wagner (Min-cut)	8	8.4 Point In Convex Hull	20
4.7 GomoryHu Tree	8	8.5 Point In Circle	20
4.8 General Graph Matching	8	8.6 Half Plane Intersection	20
4.9 Flow notes	9	8.7 Minkowski Sum	20
5 String	9	8.8 Polar Angle	20
5.1 AC Automaton	9	8.9 Rotating Sweep Line	21
5.2 Lyndon Factorization	10	8.10 Segment Intersect	21
5.3 KMP	10	8.11 Circle Intersect With Any	21
5.4 Manacher	10	8.12 Tangents	21
5.5 Minimum Rotate	10	8.13 Tangent to Convex Hull	22
5.6 Palindrome Tree	10	8.14 Minimum Enclosing Circle	22
5.7 Palindrome Partition	10	8.15 Union of Stuff	22
5.8 Repetition	11	8.16 Delaunay Triangulation	22
5.9 Suffix Array	11	8.17 Voronoi Diagram	23
5.10 SAIS (C++20)	11	8.18 3D Basic	23
5.11 Suffix Automaton	12	8.19 3D Convex Hull	23
5.12 Z Value	12	9 Misc	24
		9.1 All LCS	24
		9.2 Binary Search On Fraction	24
		9.3 Cyclic Ternary Search	24
		9.4 Matroid Intersection	24
		9.5 Min Plus Convolution	24
		9.6 Mo's Algorithm	24
		9.7 Mo's Algorithm On Tree	24
		9.8 PBDS	25
		9.9 Simulated Annealing	25
		9.10 SOS dp	25
		9.11 SMAWK	25
		9.12 Tree Hash	25
		9.13 Python	25

1 Basic

1.1 Default Code [81fabd]

```
#include <bits/stdc++.h>
#define For(i, a, b) for (int i = a; i <= b; i++)
#define Forr(i, a, b) for (int i = a; i >= b; i--)
#define F first
#define S second
#define fs F
#define sc S
#define iter(x) x.begin(), x.end()
#define sz(x) ((int)x.size())
using namespace std;
using ll = long long;
using pii = pair<int, int>;
using pll = pair<ll, ll>;

int32_t main() {
    ios::sync_with_stdio(false);
    cin.tie(0); // ^= -w -^=
}
```

1.2 Debugger [0d7909]

```
#ifdef zisk
template<class T> void _d(T &&x) {
    if constexpr (ranges::range<T> &&
        !is_convertible_v<T, string_view>) {
        cerr << "{ "; for (auto &&i:x) _d(i); cerr << " } ";
    } else if constexpr (requires { get<0>(x); }) {
        cerr << "( ";
        apply([&](auto &&...a){ (_d(a), ...); }, x);
        cerr << " ) ";
    } else cerr << x << " ";
} // ranges::subrange(L, r)
void _db() { cerr << "\e[0m\n"; }
void _db(auto &&...a) {
    cerr << "\e[1;33m"; (_d(a), ...); _db(); }
#define debug(...) _db(#__VA_ARGS__, ":", __VA_ARGS__)
#define safe_db(__PRETTY_FUNCTION__, __LINE__, "safe")
#else
#define safe void()
#define debug(...) void()
#endif
```

1.3 Fast IO [604966]

```
#pragma GCC optimize("Ofast,inline,unroll-loops")
#pragma GCC target("bmi,bmi2,lzcnt,popcnt,avx2,sse,sse2,sse3,ssse3,sse4,popcnt,abm,mmx,avx,tune=native")
#include<unistd.h>
char OB[65536]; int OP;
inline char RC() {
    static char buf[65536], *p = buf, *q = buf;
    return p == q && (q = (p = buf) + read(0, buf, 65536))
        == buf ? -1 : *p++;
}
inline int R() {
    static char c;
    while((c = RC()) < '0'); int a = c ^ '0';
    while((c = RC()) >= '0') a *= 10, a += c ^ '0';
    return a;
}
inline void W(int n) {
    static char buf[12], p;
    if (n == 0) OB[OP++] = '0'; p = 0;
    while (n) buf[p++] = '0' + (n % 10), n /= 10;
    for (--p; p >= 0; --p) OB[OP++] = buf[p];
    if (OP > 65520) write(1, OB, OP), OP = 0;
}
```

1.4 vimrc [e7a474]

```
sy on
se nu rnu so=8 bs=2 sw=4 ts=4 sts=4 hls ls=2 cin acd
mouse=a et
map <F9> :w<bar>!g++ "%" -o %:r -std=c++20 -Wall -
Wextra -Wshadow -Wconversion -O2 -Dzisk -g -
fsanitize=address,undefined<CR>
map <F8> :!./%:r<CR>
inoremap {<CR> {<CR>}<ESC>O
ca Hash w !cpp -dD -P -fpreprocessed \ tr -d '[:space
:]' \ md5sum \ cut -c-6
inoremap fj <ESC>
vnoremap fj <ESC>
// -D_GLIBCXX_ASSERTIONS, -D_GLIBCXX_DEBUG
```

2 Graph

2.1 SCC [79ed0d]

```
int BuildScc(int n, const auto &adj, const auto &radj,
    auto &scc) { // 0-indexed, returns # of SCCs
    vector<int> stk;
    int cnt = 0;
    auto dfs1 = [&](auto &self, int x) -> void {
        scc[x] = -1;
        for (auto i : adj[x])
            if (scc[i] == 0) self(self, i);
        stk.emplace_back(x);
    };
    auto dfs2 = [&](auto &self, int x) -> void {
        scc[x] = cnt;
        for (auto i : radj[x])
            if (scc[i] == -1) self(self, i);
    };
    for (int i = 0; i < n; i++)
        if (scc[i] == -1) dfs1(self, i);
    for (int i = 0; i < n; i++)
        if (scc[i] == -1) dfs2(self, i);
    return cnt;
}
```

```

};
fill(scc, scc + n, 0);
For(i, 0, n - 1) if (scc[i] == 0) dfs1(dfs1, i);
for (auto i : stk | views::reverse)
    if (scc[i] == -1) dfs2(dfs2, i), cnt++;
return cnt;
}
// 2-SAT: alloc 2n vertices x0, x1
// x or y .. x0 -> y1, y0 -> x1
// forced x .. x0 -> x1
// no sol if scc[x0] == scc[x1]
// else, assign with the one with larger SCC id

```

2.2 BCC [f1ccc5]

```

struct BCC { // 0-based, allow multi edges but not allow
    loops
    int n, m, cnt = 0;
    // n:|V|, m:|E|, cnt:#bcc
    // bcc i : vertices bcc_v[i] and edges bcc_e[i]
    vector<vector<int>> bcc_v, bcc_e;
    vector<vector<pii>> g; // original graph
    vector<pii> edges; // 0-based
    BCC(int _n, vector<pii> _edges):
        n(_n), m(sz(_edges)), g(_n), edges(_edges){
        for(int i = 0; i < m; i++){
            auto [u, v] = edges[i];
            g[u].pb(pii(v, i)); g[v].pb(pii(u, i));
        }
    }
    void make_bcc(){ bcc_v.pb(); bcc_e.pb(); cnt++; }
    // modify these if you need more information
    void add_v(int v){ bcc_v.back().pb(v); }
    void add_e(int e){ bcc_e.back().pb(e); }
    void build(){
        vector<int> dfn(n, -1), low(n, -1), stk;
        vector<vector<int>> up(n);
        int ts = 0;
        auto _dfs = [&](auto dfs, int now, int par, int pe)
            -> void{
            if(pe != -1) up[now].pb(pe);
            in[now] = low[now] = ts++;
            stk.pb(now);
            for(auto [v, e] : g[now]){
                if(e == pe) continue;
                if(in[v] != -1){
                    if(in[v] < in[now]) up[now].pb(e);
                    low[now] = min(low[now], in[v]);
                    continue;
                }
                dfs(dfs, v, now, e);
                low[now] = min(low[now], low[v]);
            } // For Edge BCC, replace to low[now] == in[now]
                (for any now)
            if((now != par && low[now] >= in[par]) || (now ==
                par && sz(g[now]) == 0)){
                make_bcc();
                for(int v = stk.back(); v = stk.back()){
                    stk.pop_back(), add_v(v);
                    for(int e : up[v]) add_e(e);
                    if(v == now) break;
                }
                if(now != par) add_v(par);
            }
        };
        for(int i = 0; i < n; i++)
            if(in[i] == -1) _dfs(_dfs, i, i, -1);
    }
};

```

2.3 Centroid Decomposition [77ce50]

```

struct CenDecomp {
    int n, mxdep;
    vector<int> par, dep;
    CenDecomp(int _n, const auto &adj): n(_n), mxdep(0),
        par(n, -1), dep(n, -1) {
        vector<pii> szs(n); int ns = 0;
        auto getsz = [&](auto &self, int v, int p) -> int {
            int sz = 1;
            for (auto i : adj[v]) if (i != p && dep[i] == -1)
                sz += self(self, i, v);
        };
    }
};

```

```

        szs[ns++] = {sz, v};
        return sz;
    };
    auto dfs = [&](auto &self, int v, int p, int d) ->
        void {
            int tot = getsz(getsz, v, v), j = 0;
            while (szs[j].F <= tot / 2) j++;
            v = szs[j].S; ns = 0;
            par[v] = p; dep[v] = d;
            mxdep = max(mxdep, d + 1);
            for (auto i : adj[v]) if (dep[i] == -1)
                self(self, i, v, d + 1);
        };
    dfs(dfs, 0, -1, 0);
} // adj could be vector<vector> or vector[]
}; // 0-indexed nodes, par[root] = -1, dep[root] = 0

```

2.4 Count Cycles [c7e8f2]

```

// ord = sort by deg decreasing, rk[ord[i]] = i
// D[i] = edge point from rk small to rk big
for (int x : ord) { // c3
    for (int y : D[x]) vis[y] = 1;
    for (int y : D[x]) for (int z : D[y]) c3 += vis[z];
    for (int y : D[x]) vis[y] = 0;
}
for (int x : ord) { // c4
    for (int y : D[x]) for (int z : adj[y])
        if (rk[z] > rk[x]) c4 += vis[z]++;
    for (int y : D[x]) for (int z : adj[y])
        if (rk[z] > rk[x]) --vis[z];
} // both are O(M*sqrt(M)), test @ 2022 CCPC guangzhou

```

2.5 DirectedMST [d3eb5f]

```

using D = int;
struct edge {
    int u, v; D w;
};
// 0-based, return index of edges
vector<int> dmst(vector<edge> &e, int n, int root) {
    using T = pair<D, int>;
    using PQ = pair<priority_queue<T, vector<T>,
        greater<T>>, D>;
    auto push = [](PQ &pq, T v) {
        pq.first.emplace(v.first - pq.second, v.second);
    };
    auto top = [](const PQ &pq) -> T {
        auto r = pq.first.top();
        return {r.first + pq.second, r.second};
    };
    auto join = [&push, &top](PQ &a, PQ &b) {
        if (a.first.size() < b.first.size()) swap(a, b);
        while (!b.first.empty())
            push(a, top(b)), b.first.pop();
    };
    vector<PQ> h(n * 2);
    for (int i = 0; i < e.size(); ++i)
        push(h[e[i].v], {e[i].w, i});
    vector<int> a(n * 2), v(n * 2, -1), pa(n * 2, -1), r(
        n * 2);
    iota(a.begin(), a.end(), 0);
    auto o = [&](int x) { int y;
        for (y = x; a[y] != y; y = a[y]);
        for (int ox = x; x != y; ox = x)
            x = a[x], a[ox] = y;
        return y;
    };
    v[root] = n + 1;
    int pc = n;
    for (int i = 0; i < n; ++i) if (v[i] == -1) {
        for (int p = i; v[p] == -1 || v[p] == i; p = o(e[r[
            p]].u)) {
            if (v[p] == i) {
                int q = p; p = pc++;
                do {
                    h[q].second = -h[q].first.top().first;
                    join(h[pa[q]] = a[q] = p, h[q]);
                } while ((q = o(e[r[q]].u)) != p);
            }
            v[p] = i;
            while (!h[p].first.empty() && o(e[top(h[p]).
                second].u) == p)
        }
    }
}

```

```

        h[p].first.pop();
        r[p] = top(h[p]).second;
    }
}
vector<int> ans;
for (int i = pc - 1; i >= 0; i--)
    if (i != root && v[i] != n) {
        for (int f = e[r[i]].v; f != -1 && v[f] != n; f =
            pa[f]) v[f] = n;
        ans.pb(r[i]);
    }
return ans;
}

```

2.6 Dominator Tree [6378d5]

```

struct Dominator_tree {
    int n, id;
    vector<vector<int>> adj, radj, bucket;
    vector<int> sdom, dom, vis, rev, par, rt, mn;
    Dominator_tree (int _n) : n(_n), id(0) {
        adj.resize(n), radj.resize(n), bucket.resize(n);
        sdom.resize(n), dom.resize(n, -1), vis.resize(n,
            -1);
        rev.resize(n), rt.resize(n), mn.resize(n), par.
            resize(n);
    }
    void add_edge(int u, int v) {adj[u].pb(v);}
    int query(int v, bool x) {
        if (rt[v] == v) return x ? -1 : v;
        int p = query(rt[v], true);
        if (p == -1) return x ? rt[v] : mn[v];
        if (sdom[mn[v]] > sdom[mn[rt[v]]]) mn[v] = mn[rt[v]
            ];
        rt[v] = p;
        return x ? p : mn[v];
    }
    void dfs(int v) {
        vis[v] = id, rev[id] = v;
        rt[id] = mn[id] = sdom[id] = id, id++;
        for (int u : adj[v]) {
            if (vis[u] == -1) dfs(u), par[vis[u]] = vis[v];
            radj[vis[u]].pb(vis[v]);
        }
    }
    void build(int s) {
        dfs(s);
        for (int i = id - 1; ~i; --i) {
            for (int u : radj[i]) {
                sdom[i] = min(sdom[i], sdom[query(u, false)]);
            }
            if (i) bucket[sdom[i]].pb(i);
            for (int u : bucket[i]) {
                int p = query(u, false);
                dom[u] = sdom[p] == i ? i : p;
            }
            if (i) rt[i] = par[i];
        }
        vector<int> res(n, -1);
        for (int i = 1; i < id; ++i) {
            if (dom[i] != sdom[i]) dom[i] = dom[dom[i]];
        }
        for (int i = 1; i < id; ++i) res[rev[i]] = rev[dom[
            i]];
        res[s] = s;
        dom = res;
    } // dom[]: parent on dom. tree, -1 if not reachable
};

```

2.7 Heavy Light Decomposition [372958]

```

vector<int> adj[N];
struct HLD {
    // 0-index
    vector<int> dep, pt, hd, idx, sz, par, vis;
    int n, _t;
    HLD () = default;
    HLD (int _n) : n(_n) {
        pt.assign(n, -1), hd.assign(n, -1), par.assign(n, -1);
        idx.assign(n, 0), sz.assign(n, 0), dep.assign(n, 0),
            vis.assign(n, 0);
        _t = 0;
    }
}

```

```

for (int i = 0; i < n; ++i) if (!vis[i]) {
    dfs1(i, -1);
    dfs2(i, -1, 0);
}
}
void dfs1(int v, int pa) {
    par[v] = pa;
    dep[v] = ~pa ? dep[pa] + 1 : 0;
    sz[v] = vis[v] = 1;
    for (int u : adj[v]) if (u != pa) {
        dfs1(u, v);
        if (pt[v] == -1 || sz[pt[v]] < sz[u])
            pt[v] = u;
        sz[v] += sz[u];
    }
}
void dfs2(int v, int pa, int h) {
    if (v == -1)
        return;
    idx[v] = _t++, hd[v] = h;
    dfs2(pt[v], v, h);
    for (int u : adj[v]) if (u != pa && u != pt[v]) {
        dfs2(u, v, u);
    }
}
void modify(int u, int v) {
    while (hd[u] != hd[v]) {
        if (dep[hd[u]] < dep[hd[v]])
            swap(u, v);
        // range [idx[hd[u]], idx[u] + 1)
        u = par[hd[u]];
    }
    if (dep[u] < dep[v])
        swap(u, v);
    // range [idx[v], idx[u] + 1)
}
int query(int u, int v) {
    int ans = 0;
    while (hd[u] != hd[v]) {
        if (dep[hd[u]] < dep[hd[v]])
            swap(u, v);
        // range [idx[hd[u]], idx[u] + 1)
        u = par[hd[u]];
    }
    if (dep[u] < dep[v])
        swap(u, v);
    // range [idx[v], idx[u] + 1)
    return ans;
}
};

```

2.8 Virtual Tree [dcbe4f]

```

// need lca
vector<int> _g[N], stk;
int st[N], ed[N];
void solve(vector<int> v) {
    auto cmp = [&](int x, int y) {return st[x] < st[y];};
    sort(all(v), cmp);
    int sz = v.size();
    for (int i = 0; i < sz - 1; ++i)
        v.pb(lca(v[i], v[i + 1]));
    sort(all(v), cmp);
    v.resize(unique(all(v)) - v.begin());
    stk.clear(), stk.pb(v[0]);
    for (int i = 1; i < v.size(); ++i) {
        int x = v[i];
        while (ed[stk.back()] < ed[x]) stk.pop_back();
        _g[stk.back()].pb(x), stk.pb(x);
    }
    // do something
    for (int i : v) _g[i].clear();
}

```

2.9 Vizing [fa4b32]

```

namespace vizing { // returns edge coloring in adjacent
    matrix G, 1 - based
    const int N = 105;
    int C[N][N], G[N][N], X[N], vst[N], n;
    void init(int _n) { n = _n;
        for (int i = 0; i <= n; ++i)
    }
}

```

```

    for (int j = 0; j <= n; ++j)
        C[i][j] = G[i][j] = 0;
}
void solve(vector<pii> &E) {
    auto update = [&](int u)
    { for (X[u] = 1; C[u][X[u]]; ++X[u]); };
    auto color = [&](int u, int v, int c) {
        int p = G[u][v];
        G[u][v] = G[v][u] = c;
        C[u][c] = v, C[v][c] = u;
        C[u][p] = C[v][p] = 0;
        if (p) X[u] = X[v] = p;
        else update(u), update(v);
        return p;
    };
    auto flip = [&](int u, int c1, int c2) {
        int p = C[u][c1];
        swap(C[u][c1], C[u][c2]);
        if (p) G[u][p] = G[p][u] = c2;
        if (!C[u][c1]) X[u] = c1;
        if (!C[u][c2]) X[u] = c2;
        return p;
    };
    fill_n(X + 1, n, 1);
    for (int t = 0; t < E.size(); ++t) {
        int u = E[t].F, v0 = E[t].S, v = v0, c0 = X[u], c =
            c0, d;
        vector<pii> L;
        fill_n(vst + 1, n, 0);
        while (!G[u][v0]) {
            L.emplace_back(v, d = X[v]);
            if (!C[v][c]) for (int a = (int)L.size() - 1; a
                >= 0; --a) c = color(u, L[a].F, c);
            else if (!C[u][d]) for (int a = (int)L.size() -
                1; a >= 0; --a) color(u, L[a].F, L[a].S);
            else if (vst[d]) break;
            else vst[d] = 1, v = C[u][d];
        }
        if (!G[u][v0]) {
            for (; v; v = flip(v, c, d), swap(c, d));
            if (int a; C[u][c0]) {
                for (a = (int)L.size() - 2; a >= 0 && L[a].S !=
                    c; --a);
                for (; a >= 0; --a) color(u, L[a].F, L[a].S);
            }
            else --t;
        }
    }
}
// namespace vizing

```

2.10 Maximum Clique Dynamic [6dde09]

```

const int N = 150;
struct MaxClique { // Maximum Clique
    bitset<N> a[N], cs[N];
    int ans, sol[N], q, cur[N], d[N], n;
    void init(int _n) {
        n = _n;
        for (int i = 0; i < n; i++) a[i].reset();
    }
    void addEdge(int u, int v) { a[u][v] = a[v][u] = 1; }
    void csort(vector<int> &r, vector<int> &c) {
        int mx = 1, km = max(ans - q + 1, 1), t = 0,
            m = r.size();
        cs[1].reset(), cs[2].reset();
        for (int i = 0; i < m; i++) {
            int p = r[i], k = 1;
            while ((cs[k] & a[p]).count()) k++;
            if (k > mx) mx++, cs[mx + 1].reset();
            cs[k][p] = 1;
            if (k < km) r[t++] = p;
        }
        c.resize(m);
        if (t) c[t - 1] = 0;
        for (int k = km; k <= mx; k++)
            for (int p = cs[k]._Find_first(); p < N;
                p = cs[k]._Find_next(p))
                r[t] = p, c[t] = k, t++;
    }
    void dfs(vector<int> &r, vector<int> &c, int l,
        bitset<N> mask) {

```

```

        while (!r.empty()) {
            int p = r.back();
            r.pop_back(), mask[p] = 0;
            if (q + c.back() <= ans) return;
            cur[q++] = p;
            vector<int> nr, nc;
            bitset<N> nmask = mask & a[p];
            for (int i : r)
                if (a[p][i]) nr.push_back(i);
            if (!nr.empty()) {
                if (1 < 4) {
                    for (int i : nr)
                        d[i] = (a[i] & nmask).count();
                    sort(nr.begin(), nr.end(),
                        [&](int x, int y) { return d[x] > d[y]; });
                }
                csort(nr, nc), dfs(nr, nc, l + 1, nmask);
            } else if (q > ans) ans = q, copy_n(cur, q, sol);
            c.pop_back(), q--;
        }
    }
    int solve(bitset<N> mask = bitset<N>(
        string(N, '1'))) { // vertex mask
        vector<int> r, c;
        ans = q = 0;
        for (int i = 0; i < n; i++)
            if (mask[i]) r.push_back(i);
        for (int i = 0; i < n; i++)
            d[i] = (a[i] & mask).count();
        sort(r.begin(), r.end(),
            [&](int i, int j) { return d[i] > d[j]; });
        csort(r, c), dfs(r, c, 1, mask);
        return ans; // sol[0 ~ ans-1]
    }
} graph;

```

2.11 Minimum Steiner Tree [c87e6f]

```

// O(V 3^T + V^2 2^T)
struct SteinerTree { // 0-base
    static const int T = 10, N = 105, INF = 1e9;
    int n, dst[N][N], dp[1 << T][N], tdst[N];
    int vcost[N]; // the cost of vertices
    void init(int _n) {
        n = _n;
        for (int i = 0; i < n; ++i) {
            for (int j = 0; j < n; ++j) dst[i][j] = INF;
            dst[i][i] = vcost[i] = 0;
        }
    }
    void add_edge(int ui, int vi, int wi) {
        dst[ui][vi] = min(dst[ui][vi], wi);
    }
    void shortest_path() {
        for (int k = 0; k < n; ++k)
            for (int i = 0; i < n; ++i)
                for (int j = 0; j < n; ++j)
                    dst[i][j] =
                        min(dst[i][j], dst[i][k] + dst[k][j]);
    }
    int solve(const vector<int> &ter) {
        shortest_path();
        int t = sz(ter);
        for (int i = 0; i < (1 << t); ++i)
            for (int j = 0; j < n; ++j) dp[i][j] = INF;
        for (int i = 0; i < n; ++i) dp[0][i] = vcost[i];
        for (int msk = 1; msk < (1 << t); ++msk) {
            if (!(msk & (msk - 1))) {
                int who = __lg(msk);
                for (int i = 0; i < n; ++i)
                    dp[msk][i] =
                        vcost[ter[who]] + dst[ter[who]][i];
            }
            for (int i = 0; i < n; ++i)
                for (int submsk = (msk - 1) & msk; submsk =
                    (submsk - 1) & msk)
                    dp[msk][i] = min(dp[msk][i],
                        dp[submsk][i] + dp[msk ^ submsk][i] -
                            vcost[i]);
            for (int i = 0; i < n; ++i) {
                tdst[i] = INF;
                for (int j = 0; j < n; ++j)

```

```

        tdst[i] =
            min(tdst[i], dp[msk][j] + dst[j][i]);
    }
    for (int i = 0; i < n; ++i) dp[msk][i] = tdst[i];
}
int ans = INF;
for (int i = 0; i < n; ++i)
    ans = min(ans, dp[(1 << t) - 1][i]);
return ans;
}
};

```

2.12 Theory

$| \text{Maximum independent edge set} | = |V| - | \text{Minimum edge cover} |$
 $| \text{Maximum independent set} | = |V| - | \text{Minimum vertex cover} |$

3 Data Structure

3.1 Dynamic Line Hull [8ec1c7]

```

struct Line {
    mutable ll k, m, p;
    bool operator<(const Line& o) const { return k < o.k; }
    bool operator<(ll x) const { return p < x; }
};

struct LineContainer : multiset<Line, less<>> {
    static const ll inf = LLONG_MAX;
    ll div(ll a, ll b) {
        return a / b - ((a ^ b) < 0 && a % b);
    }
    bool isect(iterator x, iterator y) {
        if (y == end()) return x->p = inf, 0;
        if (x->k == y->k) x->p = x->m > y->m ? inf : -inf;
        else x->p = div(y->m - x->m, x->k - y->k);
        return x->p >= y->p;
    }
    void add(ll k, ll m) {
        auto z = insert({k, m, 0}), y = z++, x = y;
        while (isect(y, z)) z = erase(z);
        if (x != begin() && isect(--x, y)) isect(x, y = erase(y));
        while ((y = x) != begin() && (--x)->p >= y->p)
            isect(x, erase(y));
    }
    ll query(ll x) {
        assert(!empty());
        auto l = *lower_bound(x);
        return l.k * x + l.m;
    }
};

```

3.2 Leftist Tree [473c12]

```

struct node {
    ll rk, data, sz, sum;
    node *l, *r;
    node(ll k) : rk(0), data(k), sz(1), l(0), r(0), sum(k) {}
};

ll sz(node *p) { return p ? p->sz : 0; }
ll rk(node *p) { return p ? p->rk : -1; }
ll sum(node *p) { return p ? p->sum : 0; }

node *merge(node *a, node *b) {
    if (!a || !b) return a ? a : b;
    if (a->data < b->data) swap(a, b);
    a->r = merge(a->r, b);
    if (rk(a->r) > rk(a->l)) swap(a->r, a->l);
    a->rk = rk(a->r) + 1, a->sz = sz(a->l) + sz(a->r) + 1;
    a->sum = sum(a->l) + sum(a->r) + a->data;
    return a;
}

void pop(node *&o) {
    node *tmp = o;
    o = merge(o->l, o->r);
    delete tmp;
}

```

3.3 Link Cut Tree [87ade4]

```

// weighted subtree size, weighted path max
struct LCT {
    int ch[N][2], pa[N], v[N], sz[N], sz2[N], w[N], mx[N], _id;
    // sz := sum of v in splay, sz2 := sum of v in virtual subtree
    // mx := max w in splay
    bool rev[N];
    LCT() : _id(1) {}
    int newnode(int _v, int _w) {
        int x = _id++;
        ch[x][0] = ch[x][1] = pa[x] = 0;
        v[x] = sz[x] = _v;
        sz2[x] = 0;
        w[x] = mx[x] = _w;
        rev[x] = false;
        return x;
    }
    void pull(int i) {
        sz[i] = v[i] + sz2[i];
        mx[i] = w[i];
        if (ch[i][0])
            sz[i] += sz[ch[i][0]], mx[i] = max(mx[i], mx[ch[i][0]]);
        if (ch[i][1])
            sz[i] += sz[ch[i][1]], mx[i] = max(mx[i], mx[ch[i][1]]);
    }
    void push(int i) {
        if (rev[i]) reverse(ch[i][0]), reverse(ch[i][1]),
            rev[i] = false;
    }
    void reverse(int i) {
        if (!i) return;
        swap(ch[i][0], ch[i][1]);
        rev[i] ^= true;
    }
    bool isrt(int i) { // rt of splay
        if (!pa[i]) return true;
        return ch[pa[i]][0] != i && ch[pa[i]][1] != i;
    }
    void rotate(int i) {
        int p = pa[i], x = ch[p][1] == i, c = ch[i][!x], gp = pa[p];
        if (ch[gp][0] == p) ch[gp][0] = i;
        else if (ch[gp][1] == p) ch[gp][1] = i;
        pa[i] = gp, ch[i][!x] = p, pa[p] = i;
        ch[p][x] = c, pa[c] = p;
        pull(p), pull(i);
    }
    void splay(int i) {
        vector<int> anc;
        anc.push_back(i);
        while (!isrt(anc.back())) anc.push_back(pa[anc.back()]);
        while (!anc.empty()) push(anc.back()), anc.pop_back();
        while (!isrt(i)) {
            int p = pa[i];
            if (!isrt(p)) rotate(ch[p][1] == i ^ ch[pa[p]][1] == p ? i : p);
            rotate(i);
        }
    }
    void access(int i) {
        int last = 0;
        while (i) {
            splay(i);
            if (ch[i][1])
                sz2[i] += sz[ch[i][1]];
            sz2[i] -= sz[last];
            ch[i][1] = last;
            pull(i), last = i, i = pa[i];
        }
    }
    void makert(int i) {
        access(i), splay(i), reverse(i);
    }
    void link(int i, int j) {
        // assert(findrt(i) != findrt(j));
    }
}

```

```

    makert(i);
    makert(j);
    pa[i] = j;
    sz2[j] += sz[i];
    pull(j);
}
void cut(int i, int j) {
    makert(i), access(j), splay(i);
    // assert(sz[i] == 2 && ch[i][1] == j);
    ch[i][1] = pa[j] = 0, pull(i);
}
int findrt(int i) {
    access(i), splay(i);
    while (ch[i][0]) push(i), i = ch[i][0];
    splay(i);
    return i;
}
};

```

3.4 Splay Tree [e9029a]

```

struct Splay {
    int pa[N], ch[N][2], sz[N], rt, _id;
    ll v[N];
    Splay() {}
    void init() {
        rt = 0, pa[0] = ch[0][0] = ch[0][1] = -1;
        sz[0] = 1, v[0] = inf;
    }
    int newNode(int p, int x) {
        int id = _id++;
        v[id] = x, pa[id] = p;
        ch[id][0] = ch[id][1] = -1, sz[id] = 1;
        return id;
    }
    void rotate(int i) {
        int p = pa[i], x = ch[p][1] == i, gp = pa[p], c =
            ch[i][!x];
        sz[p] -= sz[i], sz[i] += sz[p];
        if (~c) sz[p] += sz[c], pa[c] = p;
        ch[p][x] = c, pa[p] = i;
        pa[i] = gp, ch[i][!x] = p;
        if (~gp) ch[gp][ch[gp][1] == p] = i;
    }
    void splay(int i) {
        while (~pa[i]) {
            int p = pa[i];
            if (~pa[p]) rotate(ch[pa[p]][1] == p ^ ch[p][1]
                == i ? i : p);
            rotate(i);
        }
        rt = i;
    }
    int lower_bound(int x) {
        int i = rt, last = -1;
        while (true) {
            if (v[i] == x) return splay(i), i;
            if (v[i] > x) {
                last = i;
                if (ch[i][0] == -1) break;
                i = ch[i][0];
            }
            else {
                if (ch[i][1] == -1) break;
                i = ch[i][1];
            }
        }
        splay(i);
        return last; // -1 if not found
    }
    void insert(int x) {
        int i = lower_bound(x);
        if (i == -1) {
            // assert(ch[rt][1] == -1);
            int id = newNode(rt, x);
            ch[rt][1] = id, ++sz[rt];
            splay(id);
        }
        else if (v[i] != x) {
            splay(i);
            int id = newNode(rt, x), c = ch[rt][0];
            ch[rt][0] = id;
        }
    }
};

```

```

        ch[id][0] = c;
        if (~c) pa[c] = id, sz[id] += sz[c];
        ++sz[rt];
        splay(id);
    }
};

```

4 Flow/Matching

4.1 Hopcroft Karp [4b930f]

```

struct HopcroftKarp {
    const int INF = 1 << 30;
    vector<int> adj[N];
    int match[N], dis[N], v, n, m;
    bool matched[N], vis[N];
    bool dfs(int x) { // dffdc
        vis[x] = true;
        for (int y : adj[x])
            if (match[y] == -1 || (dis[match[y]] == dis[x] +
                1 && !vis[match[y]] && dfs(match[y]))) {
                match[y] = x, matched[x] = true;
                return true;
            }
        return false;
    }
    bool bfs() { // c44dc
        memset(dis, -1, sizeof(int) * n);
        queue<int> q;
        for (int x = 0; x < n; ++x) if (!matched[x])
            dis[x] = 0, q.push(x);
        int mx = INF;
        while (!q.empty()) {
            int x = q.front(); q.pop();
            for (int y : adj[x]) {
                if (match[y] == -1) {
                    mx = dis[x];
                    break;
                }
                else if (dis[match[y]] == -1)
                    dis[match[y]] = dis[x] + 1, q.push(match[y]);
            }
        }
        return mx < INF;
    }
    int solve() { // 39e2af
        int res = 0;
        memset(match, -1, sizeof(int) * m);
        memset(matched, 0, sizeof(bool) * n);
        while (bfs()) {
            memset(vis, 0, sizeof(bool) * n);
            for (int x = 0; x < n; ++x) if (!matched[x])
                res += dfs(x);
        }
        return res;
    }
    void init(int _n, int _m) {
        n = _n, m = _m;
        for (int i = 0; i < n; ++i) adj[i].clear();
    }
    void add_edge(int x, int y) {
        adj[x].pb(y);
    }
};

```

4.2 Dinic [aa8522]

```

template <typename T> // 0-based,  $O(V^2E)$ , unit flow:  $O(\min(V^{2/3}E, E^{3/2}))$ , bipartite matching:  $O(\sqrt{VE})$ 
struct Dinic { // 0-based
    const T INF = numeric_limits<T>::max() / 2;
    struct edge { int to, rev; T cap, flow; };
    int n, s, t;
    vector<vector<edge>> g;
    vector<int> dis, cur;
    T dfs(int u, T cap) { // 09625c
        if (u == t || !cap) return cap;
        for (int &i = cur[u]; i < (int)g[u].size(); ++i) {
            edge &e = g[u][i];
            if (dis[e.to] == dis[u] + 1 && e.flow != e.cap) {

```

```

    T df = dfs(e.to, min(e.cap - e.flow, cap));
    if (df) {
        e.flow += df;
        g[e.to][e.rev].flow -= df;
        return df;
    } } }
    dis[u] = -1;
    return 0;
}
bool bfs() { // b06b3d
    fill(all(dis), -1);
    queue<int> q;
    q.push(s); dis[s] = 0;
    while (!q.empty()) {
        int v = q.front(); q.pop();
        for (auto &u : g[v])
            if (dis[u.to] == -1 && u.flow != u.cap) {
                q.push(u.to);
                dis[u.to] = dis[v] + 1;
            }
    }
    return dis[t] != -1;
}
T solve(int _s, int _t) { // 864a07
    s = _s; t = _t;
    T flow = 0, df;
    while (bfs()) {
        fill(all(cur), 0);
        while ((df = dfs(s, INF))) flow += df;
    }
    return flow;
}
void reset() {
    for (int i = 0; i < n; ++i)
        for (auto &j : g[i]) j.flow = 0;
}
void add_edge(int u, int v, T cap) {
    g[u].pb(edge{v, (int)g[v].size(), cap, 0});
    g[v].pb(edge{u, (int)g[u].size() - 1, 0, 0});
    //change g[v] to cap for undirected graphs
}
Dinic (int _n) : n(_n), g(n), dis(n), cur(n) {}
};

```

4.3 Min Cost Max Flow [e18ab8]

```

struct MCMF {
    const int INF = 1 << 30;
    struct edge {
        int v, f, c;
        edge(int _v, int _f, int _c) : v(_v), f(_f), c(_c) {}
    };
    vector<edge> E;
    vector<vector<int>> adj;
    vector<int> dis, pot, rt;
    int n, s, t;
    MCMF(int _n, int _s, int _t) : n(_n), s(_s), t(_t) {
        adj.resize(n);
    }
    void add_edge(int u, int v, int f, int c) {
        adj[u].pb(E.size(), E.pb(edge(v, f, c)));
        adj[v].pb(E.size(), E.pb(edge(u, 0, -c)));
    }
    bool SPFA() { // e9eac4
        rt.assign(n, -1), dis.assign(n, INF);
        vector<bool> vis(n, false);
        queue<int> q;
        q.push(s), dis[s] = 0, vis[s] = true;
        while (!q.empty()) {
            int v = q.front(); q.pop();
            vis[v] = false;
            for (int id : adj[v]) if (E[id].f > 0 && dis[E[id].v] > dis[v] + E[id].c + pot[v] - pot[E[id].v]) {
                dis[E[id].v] = dis[v] + E[id].c + pot[v] - pot[E[id].v];
                E[id].v, rt[E[id].v] = id;
                if (!vis[E[id].v]) vis[E[id].v] = true, q.push(E[id].v);
            }
        }
        return dis[t] != INF;
    }
};

```

```

}
bool dijkstra() { // ce8d3f
    rt.assign(n, -1), dis.assign(n, INF);
    priority_queue<pair<int, int>, vector<pair<int, int>>, greater<pair<int, int>>> pq;
    dis[s] = 0, pq.emplace(dis[s], s);
    while (!pq.empty()) {
        int d, v; tie(d, v) = pq.top(); pq.pop();
        if (dis[v] < d) continue;
        for (int id : adj[v]) if (E[id].f > 0 && dis[E[id].v] > dis[v] + E[id].c + pot[v] - pot[E[id].v]) {
            dis[E[id].v] = dis[v] + E[id].c + pot[v] - pot[E[id].v];
            E[id].v, rt[E[id].v] = id;
            pq.emplace(dis[E[id].v], E[id].v);
        }
    }
    return dis[t] != INF;
}
pair<int, int> runFlow() { // 9d16df
    pot.assign(n, 0);
    int cost = 0, flow = 0;
    bool fr = true;
    while ((fr ? SPFA() : dijkstra())) {
        for (int i = 0; i < n; ++i) {
            dis[i] += pot[i] - pot[s];
        }
        int add = INF;
        for (int i = t; i != s; i = E[rt[i] ^ 1].v) {
            add = min(add, E[rt[i]].f);
        }
        for (int i = t; i != s; i = E[rt[i] ^ 1].v) {
            E[rt[i]].f -= add, E[rt[i] ^ 1].f += add;
        }
        flow += add, cost += add * dis[t];
        fr = false;
        swap(dis, pot);
    }
    return make_pair(flow, cost);
}
};

```

4.4 Min Cost Circulation [ea0477]

```

template<typename F, typename C>
struct MinCostCirculation {
    struct ep { int to; F flow; C cost; };
    int n; vector<int> vis; int visc;
    vector<int> fa, fae; vector<vector<int>> g;
    vector<ep> e; vector<C> pi;
    MinCostCirculation(int n_) : n(n_), vis(n), visc(0),
        g(n), pi(n) {}
    void add_edge(int u, int v, F fl, C cs) {
        g[u].emplace_back((int)e.size());
        e.emplace_back(v, fl, cs);
        g[v].emplace_back((int)e.size());
        e.emplace_back(u, 0, -cs);
    }
    C phi(int x) {
        if (fa[x] == -1) return 0;
        if (vis[x] == visc) return pi[x];
        vis[x] = visc;
        return pi[x] = phi(fa[x]) - e[fae[x]].cost;
    }
    int lca(int u, int v) {
        for (; u != -1 || v != -1; swap(u, v)) if (u != -1) {
            if (vis[u] == visc) return u;
            vis[u] = visc; u = fa[u];
        }
        return -1;
    }
    void pushflow(int x, C &cost) {
        int v = e[x ^ 1].to, u = e[x].to; ++visc;
        if (int w = lca(u, v); w == -1) {
            while (v != -1)
                swap(x ^ 1, fae[v]), swap(u, fa[v]), swap(u, v);
        }
        else {
            int z = u, dir = 0; F f = e[x].flow;
            vector<int> cyc = {x};
            for (int d : {0, 1})

```

```

    for (int i = (d ? u : v); i != w; i = fa[i]) {
        cyc.push_back(fae[i] ^ d);
        if (chmin(f, e[fae[i] ^ d].flow)) z = i, dir = d;
    }
    for (int i : cyc) {
        e[i].flow -= f; e[i ^ 1].flow += f;
        cost += f * e[i].cost;
    }
    if (dir) x ^= 1, swap(u, v);
    while (u != z)
        swap(x ^= 1, fae[v]), swap(u, fa[v]), swap(u, v);
}
}
void dfs(int u) {
    vis[u] = visc;
    for (int i : g[u])
        if (int v = e[i].to; vis[v] != visc and e[i].flow)
            fa[v] = u, fae[v] = i, dfs(v);
}
C simplex() {
    fa.assign(g.size(), -1); fae.assign(g.size(), -1);
    C cost = 0; ++visc; dfs(0);
    for (int fail = 0; fail < ssize(e); )
        for (int i = 0; i < ssize(e); i++)
            if (e[i].flow and e[i].cost < phi(e[i ^ 1].to) - phi(e[i].to))
                fail = 0, pushflow(i, cost), ++visc;
            else ++fail;
    return cost;
}
};

```

4.5 Kuhn Munkres [a138de]

```

template <typename T>
struct KM { // 0-based
    const T INF = 1 << 30;
    T w[N][N], hl[N], hr[N], slk[N];
    int fl[N], fr[N], pre[N], n;
    bool vl[N], vr[N];
    queue <int> q;
    KM () {}
    void init(int _n) {
        n = _n;
        for (int i = 0; i < n; ++i)
            for (int j = 0; j < n; ++j) w[i][j] = -INF;
    }
    void add_edge(int a, int b, T wei) { w[a][b] = wei; }
    bool check(int x) {
        if (vl[x] = 1, ~fl[x])
            return q.push(fl[x]), vr[fl[x]] = 1;
        while (~x) swap(x, fr[fl[x] = pre[x]]);
        return 0;
    }
    void bfs(int s) {
        fill(slk, slk + n, INF), fill(vl, vl + n, 0);
        fill(vr, vr + n, 0);
        while (!q.empty()) q.pop();
        q.push(s), vr[s] = 1;
        while (true) {
            T d;
            while (!q.empty()) {
                int y = q.front(); q.pop();
                for (int x = 0; x < n; ++x)
                    if (!vl[x] && slk[x] >= (d = hl[x] + hr[y] - w[x][y]))
                        if (pre[x] = y, d) slk[x] = d;
                        else if (!check(x)) return;
            }
            d = INF;
            for (int x = 0; x < n; ++x)
                if (!vl[x] && d > slk[x]) d = slk[x];
            for (int x = 0; x < n; ++x) {
                if (vl[x]) hl[x] += d;
                else slk[x] -= d;
                if (vr[x]) hr[x] -= d;
            }
            for (int x = 0; x < n; ++x)
                if (!vl[x] && !slk[x] && !check(x)) return;
        }
    }
};

```

```

    }
}
T solve() {
    fill(fl, fl + n, -1), fill(fr, fr + n, -1);
    fill(hr, hr + n, 0);
    for (int i = 0; i < n; ++i)
        hl[i] = *max_element(w[i], w[i] + n);
    for (int i = 0; i < n; ++i) bfs(i);
    T res = 0;
    for (int i = 0; i < n; ++i) res += w[i][fl[i]];
    return res;
}
};

```

4.6 Stoer Wagner (Min-cut) [ac255a]

```

struct SW {
    int g[N][N], sum[N], n;
    bool vis[N], dead[N];
    void init(int _n) {
        n = _n;
        for (int i = 0; i < n; ++i) fill(g[i], g[i] + n, 0);
        fill(dead, dead + n, false);
    }
    void add_edge(int u, int v, int w) {
        g[u][v] += w, g[v][u] += w;
    }
    int run() {
        int ans = 1 << 30;
        for (int round = 0; round + 1 < n; ++round) {
            fill(vis, vis + n, false), fill(sum, sum + n, 0);
            int num = 0, s = -1, t = -1;
            while (num < n - round) {
                int now = -1;
                for (int i = 0; i < n; ++i) if (!vis[i] && !dead[i]) {
                    if (now == -1 || sum[now] < sum[i]) now = i;
                }
                s = t, t = now;
                vis[now] = true, num++;
                for (int i = 0; i < n; ++i) if (!vis[i] && !dead[i]) {
                    dead[i] = true;
                    sum[i] += g[now][i];
                }
            }
            ans = min(ans, sum[t]);
            for (int i = 0; i < n; ++i) {
                g[i][s] += g[i][t];
                g[s][i] += g[t][i];
            }
            dead[t] = true;
        }
        return ans;
    }
};

```

4.7 GomoryHu Tree [90ead2]

```

vector <array <int, 3>> GomoryHu(Dinic <int> flow) {
    // Tree edge min = mincut (0-based)
    int n = flow.n;
    vector <array <int, 3>> ans;
    vector <int> rt(n);
    for (int i = 1; i < n; ++i) {
        int t = rt[i];
        flow.reset();
        ans.pb({i, t, flow.solve(i, t)});
        flow.bfs();
        for (int j = i + 1; j < n; ++j)
            if (rt[j] == t && flow.dis[j] != -1) rt[j] = i;
    }
    return ans;
}

```

4.8 General Graph Matching [2b7f20]

```

struct Matching { // 0-based
    int fa[N], pre[N], match[N], s[N], v[N], n, tk;
    vector <int> g[N];
    queue <int> q;
    int Find(int u) {

```

```

    return u == fa[u] ? u : fa[u] = Find(fa[u]);
}
int lca(int x, int y) {
    tk++;
    x = Find(x), y = Find(y);
    for (; ; swap(x, y)) {
        if (x != n) {
            if (v[x] == tk) return x;
            v[x] = tk;
            x = Find(pre[match[x]]);
        }
    }
}
void blossom(int x, int y, int l) {
    while (Find(x) != l) {
        pre[x] = y, y = match[x];
        if (s[y] == 1) q.push(y), s[y] = 0;
        if (fa[x] == x) fa[x] = 1;
        if (fa[y] == y) fa[y] = 1;
        x = pre[y];
    }
}
bool bfs(int r) {
    for (int i = 0; i <= n; ++i) fa[i] = i, s[i] = -1;
    while (!q.empty()) q.pop();
    q.push(r);
    s[r] = 0;
    while (!q.empty()) {
        int x = q.front(); q.pop();
        for (int u : g[x]) {
            if (s[u] == -1) {
                pre[u] = x, s[u] = 1;
                if (match[u] == n) {
                    for (int a = u, b = x, last; b != n; a = last, b = pre[a])
                        last = match[b], match[b] = a, match[a] = b;
                    return true;
                }
                q.push(match[u]);
                s[match[u]] = 0;
            } else if (!s[u] && Find(u) != Find(x)) {
                int l = lca(u, x);
                blossom(x, u, l);
                blossom(u, x, l);
            }
        }
    }
    return false;
}
int solve() {
    int res = 0;
    for (int x = 0; x < n; ++x) {
        if (match[x] == n) res += bfs(x);
    }
    return res;
}
void init(int _n) {
    n = _n, tk = 0;
    for (int i = 0; i <= n; ++i) match[i] = pre[i] = n;
    for (int i = 0; i < n; ++i) g[i].clear(), v[i] = 0;
}
void add_edge(int u, int v) {
    g[u].push_back(v), g[v].push_back(u);
}
};

```

4.9 Flow notes

- Bipartite Matching Restore Answer
runBfs(); Answer is $\{vis[x]|x \in L\} \cup \{vis[x]|x \in R\}$
- Bipartite Minimum Weight Vertex Covering
 $S \rightarrow \{x|x \in L\}$, cap = weight of vertex x
 $\{x|x \in L\} \rightarrow \{y|y \in R\}$, cap = ∞
 $\{y|y \in R\} \rightarrow T$, cap = weight of vertex y
 For general version, change Dinic to MCMF and:
 $S \rightarrow \{x|x \in L\}$, cap = weight of vertex x , cost = 0
 $\{x|x \in L\} \rightarrow \{y|y \in R\}$, cap = ∞ , cost = $-w$
 $\{y|y \in R\} \rightarrow T$, cap = weight of vertex y , cost = 0
- Useful Lemma
 (Bipartite Maximum Weight Independent Set) +
 (Bipartite Minimum Weight Vertex Covering) = weight sum

- Min Cut Model
 choose A but not choose B cost $x: A \rightarrow B$, cap = x
 choose A cost $x: A \rightarrow T$, cap = x
 not choose A cost $x: S \rightarrow A$, cap = x
 choose A gain $x \implies$ not choose A cost x , tot+ = x
 choose A and choose B cost x : NO!!!
 Bipartite: can flip one side
- Min Cut Restore Answer
 runBfs(); Answer is $\{vis[x]|x \in V\}$
- Maximum/Minimum flow with lower bound / Circulation problem
 - Construct super source S and sink T .
 - For each edge (x, y, l, u) , connect $x \rightarrow y$ with capacity $u - l$.
 - For each vertex v , denote by $in(v)$ the difference between the sum of incoming lower bounds and the sum of outgoing lower bounds.
 - If $in(v) > 0$, connect $S \rightarrow v$ with capacity $in(v)$, otherwise, connect $v \rightarrow T$ with capacity $-in(v)$.
 - To maximize, connect $t \rightarrow s$ with capacity ∞ (skip this in circulation problem), and let f be the maximum flow from S to T . If $f \neq \sum_{v \in V, in(v) > 0} in(v)$, there's no solution. Otherwise, the maximum flow from s to t is the answer.
 - To minimize, let f be the maximum flow from S to T . Connect $t \rightarrow s$ with capacity ∞ and let the flow from S to T be f' . If $f + f' \neq \sum_{v \in V, in(v) > 0} in(v)$, there's no solution. Otherwise, f' is the answer.
 - The solution of each edge e is $l_e + f_e$, where f_e corresponds to the flow of edge e on the graph.
- Construct minimum vertex cover from maximum matching M on bipartite graph (X, Y)
 - Redirect every edge: $y \rightarrow x$ if $(x, y) \in M$, $x \rightarrow y$ otherwise.
 - DFS from unmatched vertices in X .
 - $x \in X$ is chosen iff x is unvisited.
 - $y \in Y$ is chosen iff y is visited.
- Minimum cost cyclic flow
 - Construct super source S and sink T
 - For each edge (x, y, c) , connect $x \rightarrow y$ with $(cost, cap) = (c, 1)$ if $c > 0$, otherwise connect $y \rightarrow x$ with $(cost, cap) = (-c, 1)$
 - For each edge with $c < 0$, sum these cost as K , then increase $d(y)$ by 1, decrease $d(x)$ by 1
 - For each vertex v with $d(v) > 0$, connect $S \rightarrow v$ with $(cost, cap) = (0, d(v))$
 - For each vertex v with $d(v) < 0$, connect $v \rightarrow T$ with $(cost, cap) = (0, -d(v))$
 - Flow from S to T , the answer is the cost of the flow $C + K$
- Maximum density induced subgraph
 - Binary search on answer, suppose we're checking answer T
 - Construct a max flow model, let K be the sum of all weights
 - Connect source $s \rightarrow v, v \in G$ with capacity K
 - For each edge (u, v, w) in G , connect $u \rightarrow v$ and $v \rightarrow u$ with capacity w
 - For $v \in G$, connect it with sink $v \rightarrow t$ with capacity $K + 2T - (\sum_{e \in E(v)} w(e)) - 2w(v)$
 - T is a valid answer if the maximum flow $f < K|V|$
- Minimum weight edge cover
 - For each $v \in V$ create a copy v' , and connect $u' \rightarrow v'$ with weight $w(u, v)$.
 - Connect $v \rightarrow v'$ with weight $2\mu(v)$, where $\mu(v)$ is the cost of the cheapest edge incident to v .
 - Find the minimum weight perfect matching on G' .
- Project selection problem
 - If $p_v > 0$, create edge (s, v) with capacity p_v ; otherwise, create edge (v, t) with capacity $-p_v$.
 - Create edge (u, v) with capacity w with w being the cost of choosing u without choosing v .
 - The mincut is equivalent to the maximum profit of a subset of projects.

5 String

5.1 AC Automaton [b9fe7c]

```

struct AC {
    int ch[N][26], to[N][26], fail[N], sz;
    vector<int> g[N];
    int cnt[N];
    AC() {sz = 0, extend();}
    void extend() {fill(ch[sz], ch[sz] + 26, 0), sz++;}
    int nxt(int u, int v) {
        if (!ch[u][v]) ch[u][v] = sz, extend();
        return ch[u][v];
    }
    int insert(string s) {
        int now = 0;
        for (char c : s) now = nxt(now, c - 'a');
        cnt[now]++;
    }
};

```

```

    return now;
}
void build_fail() {
    queue<int> q;
    for (int i = 0; i < 26; ++i) if (ch[0][i]) {
        to[0][i] = ch[0][i];
        q.push(ch[0][i]);
        g[0].push_back(ch[0][i]);
    }
    while (!q.empty()) {
        int v = q.front(); q.pop();
        for (int j = 0; j < 26; ++j) {
            to[v][j] = ch[v][j] ? ch[v][j] : to[fail[v]][j];
        }
        for (int i = 0; i < 26; ++i) if (ch[v][i]) {
            int u = ch[v][i], k = fail[v];
            while (k && !ch[k][i]) k = fail[k];
            if (ch[k][i]) k = ch[k][i];
            fail[u] = k;
            cnt[u] += cnt[k], g[k].push_back(u);
            q.push(u);
        }
    }
}
int match(string &s) {
    int now = 0, ans = 0;
    for (char c : s) {
        now = to[now][c - 'a'];
        ans += cnt[now];
    }
    return ans;
}
};

```

5.2 Lyndon Factorization [a9eeb0]

```

// partition s = w[0] + w[1] + ... + w[k-1],
// w[0] >= w[1] >= ... >= w[k-1]
// each w[i] strictly smaller than all its suffix
vector<string> duval(const string &s) {
    vector<string> ans;
    for (int n = (int)s.size(), i = 0, j, k; i < n; ) {
        for (j = i + 1, k = i; j < n && s[k] <= s[j]; j++)
            k = (s[k] < s[j] ? i : k + 1);
        for (; i <= k; i += j - k)
            ans.pb(s.substr(i, j - k)); // s.substr(L, Len)
    }
    return ans;
}

```

5.3 KMP [5ac553]

```

vector<int> build_fail(string &s) {
    vector<int> f(s.length() + 1, 0);
    int k = 0;
    for (int i = 1; i < s.length(); ++i) {
        while (k && s[k] != s[i])
            k = f[k];
        if (s[k] == s[i])
            k++;
        f[i + 1] = k;
    }
    return f;
}
int match(string &s, string &t) {
    vector<int> f = build_fail(t);
    int k = 0, ans = 0;
    for (int i = 0; i < s.length(); ++i) {
        while (k && s[i] != t[k])
            k = f[k];
        if (s[i] == t[k])
            k++;
        if (k == t.length())
            ans++, k = f[k];
    }
    return ans;
}

```

5.4 Manacher [643e55]

```
int z[MAXN]; //0-base
```

```

/* center i: radius z[i * 2 + 1] / 2
   center i, i + 1: radius z[i * 2 + 2] / 2
   both aba, abba have radius 2 */
void Manacher(string tmp) {
    string s = "%";
    int l = 0, r = 0;
    for (char c : tmp) s.pb(c), s.pb('%');
    for (int i = 0; i < s.size(); ++i) {
        z[i] = r > i ? min(z[2 * l - i], r - i) : 1;
        while (i - z[i] >= 0 && i + z[i] < s.size()
            && s[i + z[i]] == s[i - z[i]]) ++z[i];
        if (z[i] + i > r) r = z[i] + i, l = i;
    }
}

```

5.5 Minimum Rotate [acab8e]

```

string mcp(string s) {
    int n = s.size(), i = 0, j = 1;
    s += s;
    while (i < n && j < n) {
        int k = 0;
        while (k < n && s[i + k] == s[j + k]) k++;
        if (s[i + k] <= s[j + k]) j += k + 1;
        else i += k + 1;
        if (i == j) j++;
    }
    int ans = (i < n ? i : j);
    return s.substr(ans, n);
}

```

5.6 Palindrome Tree [1f78bf]

```

struct PalindromicTree {
    struct node {
        int nxt[26], fail, len; // num = depth of fail link
        int cnt, num; // cnt = occur, num = #pal_suffix of
            this node
        node(int l = 0) : nxt{}, fail(0), len(l), cnt(0), num
            (0) {}
    };
    vector<node> st; vector<int> s; int last, n;
    void init() {
        st.clear(); s.clear(); last = 1; n = 0;
        st.pb(0); st.pb(-1);
        st[0].fail = 1; s.pb(-1);
    }
    int getFail(int x) {
        while (s[n - st[x].len - 1] != s[n]) x = st[x].fail;
        return x;
    }
    void add(int c) {
        s.pb(c - 'a'); ++n;
        int cur = getFail(last);
        if (!st[cur].nxt[c]) {
            int now = sz(st);
            st.pb(st[cur].len + 2);
            st[now].fail = st[getFail(st[cur].fail)].nxt[c];
            st[cur].nxt[c] = now;
            st[now].num = st[st[now].fail].num + 1;
        }
        last = st[cur].nxt[c]; ++st[last].cnt;
    }
    void dpCnt() {
        for (int i = sz(st) - 1; i >= 0; i--) {
            auto nd = st[i];
            st[nd.fail].cnt += nd.cnt;
        }
    }
    int size() { return (int)st.size() - 2; }
};

```

5.7 Palindrome Partition [c85c05]

```

// in PAM
/* node */ int dif = 0, slink = 0, g = 0;
vector<int> dp = {0};
// add
if (!st[cur].nxt[c]) {
    // ...
    st[now].dif = st[now].len - st[st[now].fail].len;
}

```

```

if (st[now].dif == st[st[now].fail].dif)
    st[now].slink = st[st[now].fail].slink;
else st[now].slink = st[now].fail;
}
dp.pb(0);
for (int x = last; x > 1; x = st[x].slink) {
    st[x].g = dp[n - st[st[x].slink].len - st[x].dif];
    if (st[x].dif == st[st[x].fail].dif)
        st[x].g = min(st[x].g, st[st[x].fail].g);
    dp[n] = min(dp[n], st[x].g + 1);
}

```

5.8 Repetition [f3da14]

```

int to_left[N], to_right[N];
vector<array<int, 3>> rep; // L, r, Len.
// substr( [l, r], len * 2) are tandem
void findRep(string &s, int l, int r) {
    if (r - l == 1) return;
    int m = l + r >> 1;
    findRep(s, l, m), findRep(s, m, r);
    string sl = s.substr(l, m - l);
    string sr = s.substr(m, r - m);
    vector<int> Z = buildZ(sr + "#" + sl);
    for (int i = l; i < m; ++i)
        to_right[i] = Z[r - m + 1 + i - l];
    reverse(all(sl));
    Z = buildZ(sl);
    for (int i = l; i < m; ++i)
        to_left[i] = Z[m - i - l];
    reverse(all(sl));
    for (int i = l; i + 1 < m; ++i) {
        int k1 = to_left[i], k2 = to_right[i + 1];
        int len = m - i - 1;
        if (k1 < 1 || k2 < 1 || len < 2) continue;
        int tl = max(1, len - k2), tr = min(len - 1, k1);
        if (tl <= tr) rep.pb({i + 1 - tr, i + 1 - tl, len});
    }
    Z = buildZ(sr);
    for (int i = m; i < r; ++i) to_right[i] = Z[i - m];
    reverse(all(sl)), reverse(all(sr));
    Z = buildZ(sl + "#" + sr);
    for (int i = m; i < r; ++i)
        to_left[i] = Z[m - l + 1 + r - i - 1];
    reverse(all(sl)), reverse(all(sr));
    for (int i = m; i + 1 < r; ++i) {
        int k1 = to_left[i], k2 = to_right[i + 1];
        int len = i - m + 1;
        if (k1 < 1 || k2 < 1 || len < 2) continue;
        int tl = max(len - k2, 1), tr = min(len - 1, k1);
        if (tl <= tr)
            rep.pb({i + 1 - len - tr, i + 1 - len - tl, len});
    }
    Z = buildZ(sr + "#" + sl);
    for (int i = l; i < m; ++i)
        if (Z[r - m + 1 + i - l] >= m - i)
            rep.pb({i, i, m - i});
}

```

5.9 Suffix Array [4b4fc7]

```

#define FOR(i, a, b) for(int i = a; i <= b; i++)
#define ROF(i, a, b) for(int i = a; i >= b; i--)
template<int N>
struct SA {
    int sa[N], tmp[2][N], c[N], rk[N], lcp[N];
    void buildSA(const string &s) { // 94c330
        int *x = tmp[0], *y = tmp[1], n = sz(s), m = n;
        FOR(i, 0, m - 1) c[i] = 0;
        FOR(i, 0, n - 1) c[x[i]] = s[i]++;
        FOR(i, 1, m - 1) c[i] += c[i - 1];
        ROF(i, n - 1, 0) sa[--c[x[i]]] = i;
        for (int k = 1; k < n; k <= 1) {
            FOR(i, 0, m - 1) c[i] = 0;
            FOR(i, 0, n - 1) c[x[i]]++;
            FOR(i, 1, m - 1) c[i] += c[i - 1];
            int p = 0;
            FOR(i, n - k, n - 1) y[p++] = i;
            FOR(i, 0, n - 1) if (sa[i] >= k) y[p++] = sa[i] - k;
            ROF(i, n - 1, 0) sa[--c[x[y[i]]]] = y[i];
            y[sa[0]] = p = 0;

```

```

        FOR(i, 1, n - 1) {
            int a = sa[i], b = sa[i - 1];
            if (!(x[a] == x[b] && a + k < n && b + k < n &&
                x[a + k] == x[b + k])) p++;
            y[sa[i]] = p;
        }
        if (n == p + 1) break;
        swap(x, y), m = p + 1;
    }
} // sa[rk[i]] = i
void buildLCP(const string &s) { // 26131c
    int n = sz(s), val = 0;
    FOR(i, 0, n - 1) rk[sa[i]] = i;
    FOR(i, 0, n - 1) {
        if (!rk[i]) lcp[rk[i]] = 0;
        else {
            if (val) val--;
            int p = sa[rk[i] - 1];
            while (val + i < n && val + p < n && s[val + i]
                == s[val + p]) val++;
            lcp[rk[i]] = val;
        }
    }
} // lcp[i] = LCP(sa[i - 1], sa[i])
void build(const string &s) {
    buildSA(s);
    buildLCP(s);
}
};

```

5.10 SAIS (C++20) [b2c106]

```

auto sais(const auto &s) { // afa632
    const int n = sz(s), z = ranges::max(s) + 1;
    if (n == 1) return vector{0};
    vector<int> c(z); for (int x : s) ++c[x];
    partial_sum(iter(c), begin(c));
    vector<int> sa(n); auto I = views::iota(0, n);
    vector<bool> t(n, true);
    for (int i = n - 2; i >= 0; --i)
        t[i] = (s[i] == s[i + 1] ? t[i + 1] : s[i] < s[i + 1]);
    auto is_lms = views::filter([&t](int x) {
        return x && t[x] && !t[x - 1];
    });
    auto induce = [&] {
        for (auto x = c; int y : sa)
            if (y-- if (!t[y]) sa[x[s[y] - 1]++] = y;
        for (auto x = c; int y : sa | views::reverse)
            if (y-- if (t[y]) sa[--x[s[y]]] = y;
    };
    vector<int> lms, q(n); lms.reserve(n);
    for (auto x = c; int i : I | is_lms)
        q[i] = sz(lms), lms.push_back(sa[--x[s[i]]] = i);
    induce(); vector<int> ns(sz(lms));
    for (int j = -1, nz = 0; int i : sa | is_lms) {
        if (j >= 0) {
            int len = min({n - i, n - j, lms[q[i] + 1] - i});
            ns[q[i]] = nz += lexicographical_compare(
                begin(s) + j, begin(s) + j + len,
                begin(s) + i, begin(s) + i + len);
        }
        j = i;
    }
    fill(iter(sa), 0); auto nsa = sais(ns);
    for (auto x = c; int y : nsa | views::reverse)
        y = lms[y], sa[--x[s[y]]] = y;
    return induce(), sa;
}

struct Suffix {
    int n; vector<int> sa, lcp, ra;
    Suffix(const auto &s, int _n) : n(_n), lcp(n), ra(n) {
        // 86c942
        vector<int> s(n + 1); // s[n] = 0;
        copy_n(s.begin(), n, begin(s)); // _s shouldn't
        contain 0
        sa = sais(s); sa.erase(sa.begin());
        for (int i = 0; i < n; ++i) ra[sa[i]] = i;
        for (int i = 0, h = 0; i < n; ++i) {
            if (!ra[i]) { h = 0; continue; }
            for (int j = sa[ra[i] - 1]; max(i, j) + h < n &&
                s[i + h] == s[j + h];) ++h;

```

```

    lcp[ra[i]] = h ? h-- : 0;
}
} // lcp[i] = LCP(sa[i - 1], sa[i])
};

```

5.11 Suffix Automaton [277d1d]

```

struct SAM {
    int ch[N][26], len[N], link[N], pos[N], cnt[N], sz;
    // node -> strings with the same endpos set
    // length in range [len(link) + 1, len]
    // node's endpos set -> pos in the subtree of node
    // link -> longest suffix with different endpos set
    // len -> longest suffix
    // pos -> end position
    // cnt -> size of endpos set
    SAM () {len[0] = 0, link[0] = -1, pos[0] = 0, cnt[0]
        = 0, sz = 1;}
    void build(string s) {
        int last = 0;
        for (int i = 0; i < s.length(); ++i) {
            char c = s[i];
            int cur = sz++;
            len[cur] = len[last] + 1, pos[cur] = i + 1;
            int p = last;
            while (~p && !ch[p][c - 'a'])
                ch[p][c - 'a'] = cur, p = link[p];
            if (p == -1) link[cur] = 0;
            else {
                int q = ch[p][c - 'a'];
                if (len[p] + 1 == len[q]) {
                    link[cur] = q;
                } else {
                    int nxt = sz++;
                    len[nxt] = len[p] + 1, link[nxt] = link[q];
                    pos[nxt] = pos[q];
                    for (int j = 0; j < 26; ++j)
                        ch[nxt][j] = ch[q][j];
                    while (~p && ch[p][c - 'a'] == q)
                        ch[p][c - 'a'] = nxt, p = link[p];
                    link[q] = link[cur] = nxt;
                }
            }
            cnt[cur]++;
            last = cur;
        }
        vector<int> p(sz);
        iota(all(p), 0);
        sort(all(p),
            [&](int i, int j) {return len[i] > len[j];});
        for (int i = 0; i < sz; ++i)
            cnt[link[p[i]]] += cnt[p[i]];
    }
} sam;

```

5.12 Z Value [a8c33c]

```

vector<int> build(string s) {
    int n = s.length();
    vector<int> Z(n);
    int l = 0, r = 0;
    for (int i = 0; i < n; ++i) {
        Z[i] = max(min(Z[i - 1], r - i), 0);
        while (i + Z[i] < s.size() && s[Z[i]] == s[i + Z[i]]) {
            l = i, r = i + Z[i], Z[i]++;
        }
    }
    return Z;
}

```

6 Math

6.1 Characteristic Polynomial [eabd3c]

```

#define rep(x, y, z) for (int x=y; x<z; x++)
using vi = vector<int>; using vvi = vector<vi>;
void Hessenberg(vvi &H, int N) { // 0ddcb8
    for (int i = 0; i < N - 2; ++i) {
        for (int j = i + 1; j < N; ++j) if (H[j][i]) {
            rep(k, i, N) swap(H[i+1][k], H[j][k]);

```

```

            rep(k, 0, N) swap(H[k][i+1], H[k][j]);
            break;
        }
        if (!H[i + 1][i]) continue;
        for (int j = i + 2; j < N; ++j) {
            int co = mul(modinv(H[i + 1][i]), H[j][i]);
            rep(k, i, N) subeq(H[j][k], mul(H[i+1][k], co));
            rep(k, 0, N) addeq(H[k][i+1], mul(H[k][j], co));
        }
    }
}
vi CharacteristicPoly(vvi A) { // 5213a6
    int N = (int)A.size(); Hessenberg(A, N);
    vvi P(N + 1, vi(N + 1)); P[0][0] = 1;
    for (int i = 1; i <= N; ++i) {
        rep(j, 0, i+1) P[i][j] = j ? P[i-1][j-1] : 0;
        for (int j = i - 1, val = 1; j >= 0; --j) {
            int co = mul(val, A[j][i - 1]);
            rep(k, 0, j+1) subeq(P[i][k], mul(P[j][k], co));
            if (j) val = mul(val, A[j][j - 1]);
        }
    }
    if (N & 1) for (int &x: P[N]) x = sub(0, x);
    return P[N]; // test: 2021 PTZ Korea K
}

```

6.2 Discrete Logarithm [4c39b3]

```

int BSGS(int s, int x, int y, int m) { // 247dc4
    constexpr int kStep = 32000;
    unordered_map<int, int> p;
    int b = 1;
    for (int i = 0; i < kStep; ++i) {
        p[y] = i;
        y = 1ll * y * x % m;
        b = 1ll * b * x % m;
    }
    for (int i = 0; i < m + 10; i += kStep) {
        s = 1ll * s * b % m;
        if (p.find(s) != p.end()) return i + kStep - p[s];
    }
    return -1; // returns: j where s*(x^j) = y (mod m)
}
int DiscreteLog(int x, int y, int m) { // 579300
    if (m == 1) return 0;
    int s = 1;
    for (int i = 0; i < 100; ++i) {
        if (s == y) return i;
        s = 1ll * s * x % m;
    }
    if (s == y) return 100;
    int p = 100 + BSGS(s, x, y, m);
    if (fpow(x, p, m) != y) return -1;
    return p; //returns: x^p = y (mod m)
}

```

6.3 Extgcd [c51ae9]

```

// ax+ny = 1, ax+ny == ax == 1 (mod n)
void extgcd(ll x, ll y, ll &g, ll &a, ll &b) {
    if (y == 0) g = x, a = 1, b = 0;
    else extgcd(y, x % y, g, b, a), b -= (x / y) * a;
}

```

6.4 Floor Sum [49de67]

```

// sum^{n-1}_0 floor((a * i + b) / m) in Log(n + m + a + b)
ll floor_sum(ll n, ll m, ll a, ll b) {
    ll ans = 0;
    if (a >= m) ans += (n - 1) * n * (a / m) / 2, a %= m;
    if (b >= m) ans += n * (b / m), b %= m;
    ll y_max = (a * n + b) / m, x_max = (y_max * m - b);
    if (y_max == 0) return ans;
    ans += (n - (x_max + a - 1) / a) * y_max;
    ans += floor_sum(y_max, a, m, (a - x_max % a) % a);
    return ans;
}

```

6.5 Factorial Mod P^k [c324f3]

```
//  $O(p^k + \log^2 n)$ ,  $pk = p^k$ 
ll prod[MAXP];
ll fac_no_p(ll n, ll p, ll pk) {
    prod[0] = 1;
    for (int i = 1; i <= pk; ++i)
        if (i % p) prod[i] = prod[i - 1] * i % pk;
        else prod[i] = prod[i - 1];
    ll rt = 1;
    for (; n; n /= p) {
        rt = rt * mpow(prod[pk], n / pk, pk) % pk;
        rt = rt * prod[n % pk] % pk;
    }
    return rt;
} // (n! without factor p) %  $p^k$ 
```

6.6 Gaussian Elimination [0465e4]

```
using vi = vector<int>; // be careful if A.empty()
using vvi = vector<vi>; // ensure that  $0 \leq x < \text{mod}$ 
pair<vi, vvi> gauss(vvi A, vi b) { // solve  $Ax=b$ 
    const int N = (int)A.size(), M = (int)A[0].size();
    vector<int> depv, free(M, true); int rk = 0;
    for (int i = 0; i < M; i++) {
        int p = -1;
        for (int j = rk; j < N; j++)
            if (p == -1 || abs(A[j][i]) > abs(A[p][i]))
                p = j;
        if (p == -1 || A[p][i] == 0) continue;
        swap(A[p], A[rk]); swap(b[p], b[rk]);
        const int inv = modinv(A[rk][i]);
        for (int &x : A[rk]) x = mul(x, inv);
        b[rk] = mul(b[rk], inv);
        for (int j = 0; j < N; j++) if (j != rk) {
            int z = A[j][i];
            for (int k = 0; k < M; k++)
                A[j][k] = sub(A[j][k], mul(z, A[rk][k]));
            b[j] = sub(b[j], mul(z, b[rk]));
        }
        depv.push_back(i); free[i] = false; ++rk;
    }
    for (int i = rk; i < N; i++)
        if (b[i] != 0) return {{}, {}}; // not consistent
    vi x(M); vvi h;
    for (int i = 0; i < rk; i++) x[depv[i]] = b[i];
    for (int i = 0; i < M; i++) if (free[i]) {
        h.emplace_back(M); h.back()[i] = 1;
        for (int j = 0; j < rk; j++)
            h.back()[depv[j]] = sub(0, A[j][i]);
    }
    return {x, h}; // solution =  $x + \text{span}(h[i])$ 
}
```

6.7 Linear Function Mod Min [93cc4b]

```
ll topos(ll x, ll m) {x %= m; if (x < 0) x += m; return x;}
//min value of  $ax + b \pmod m$  for  $x \in [0, n - 1]$ .  $O(\log m)$ 
ll min_rem(ll n, ll m, ll a, ll b) { // 578b62
    a = topos(a, m), b = topos(b, m);
    for (ll g = __gcd(a, m); g > 1; ) return g * min_rem(n / g, a / g, b / g) + (b % g);
    for (ll nn, nm, na, nb; a; n = nn, m = nm, a = na, b = nb) {
        if (a <= m - a) {
            nn = (a * (n - 1) + b) / m;
            if (!nn) break;
            nn += (b < a);
            nm = a, na = topos(-m, a);
            nb = b < a ? b : topos(b - m, a);
        } else {
            ll lst = b - (n - 1) * (m - a);
            if (lst >= 0) {b = lst; break;}
            nn = -(lst / m) + (lst % m < -a) + 1;
            nm = m - a, na = m % (m - a), nb = b % (m - a);
        }
    }
    return b;
}
//min value of  $ax + b \pmod m$  for  $x \in [0, n - 1]$ ,
//also return min x to get the value.  $O(\log m)$ 
//{value, x}
```

```
pair<ll, ll> min_rem_pos(ll n, ll m, ll a, ll b) { //
    fd6550
    a = topos(a, m), b = topos(b, m);
    ll mn = min_rem(n, m, a, b), g, ia, ib;
    extgcd(a, m, g, ia, ib);
    //ax = (mn - b) (mod m)
    ll x = (ia + m) * ((mn - b + m) / g) % (m / g);
    return {mn, x};
}
```

6.8 MillerRabin PollardRho [9d0778]

```
// n < 4,759,123,141 3 : 2, 7, 61
// n < 1,122,004,669,633 4 : 2, 13, 23, 1662803
// n < 3,474,749,660,383 6 : primes <= 13
// n < 2^64 7 :
// 2, 325, 9375, 28178, 450775, 9780504, 1795265022
ll mul(ll a, ll b, ll n) { return (__int128)a * b % n; }
vector<ll> chk = {2, 325, 9375, 28178, 450775, 9780504, 1795265022};
ll Pow(ll a, ll b, ll n) {ll res = 1; for (; b; b >>= 1, a = mul(a, a, n)) if (b & 1) res = mul(res, a, n); return res;}
bool check(ll a, ll d, int s, ll n) { // 874cd4
    a = Pow(a, d, n);
    if (a <= 1) return 1;
    for (int i = 0; i < s; ++i, a = mul(a, a, n)) {
        if (a == 1) return 0;
        if (a == n - 1) return 1;
    }
    return 0;
}
bool IsPrime(ll n) { // b8bfee
    if (n < 2) return 0;
    if (n % 2 == 0) return n == 2;
    ll d = n - 1, s = 0;
    while (d % 2 == 0) d >>= 1, ++s;
    for (ll i : chk) if (!check(i, d, s, n)) return 0;
    return 1;
}
const vector<ll> small = {2, 3, 5, 7, 11, 13, 17, 19};
ll FindFactor(ll n) { // 697b88
    if (IsPrime(n)) return 1;
    for (ll p : small) if (n % p == 0) return p;
    ll x, y = 2, d, t = 1;
    auto f = [&](ll a) {return (mul(a, a, n) + t) % n;};
    for (int l = 2; ; l <= 1) {
        x = y;
        int m = min(1, 32);
        for (int i = 0; i < l; i += m) {
            d = 1;
            for (int j = 0; j < m; ++j) {
                y = f(y), d = mul(d, abs(x - y), n);
            }
            ll g = __gcd(d, n);
            if (g == n) {
                l = 1, y = 2, ++t;
                break;
            }
            if (g != 1) return g;
        }
    }
}
map<ll, int> PollardRho(ll n) { // 1cf996
    map<ll, int> res;
    if (n == 1) return res;
    if (IsPrime(n)) return ++res[n], res;
    ll d = FindFactor(n);
    res = PollardRho(n / d);
    auto res2 = PollardRho(d);
    for (auto [x, y] : res2) res[x] += y;
    return res;
}
```

6.9 Lucy Hedgehog [435ed1]

```
ll r, rt, ans = 0; // Goal:  $S(i, k) = \sum_{j < i, \text{lpf}(j) > p_k} f(j)$ ,  $S(i, \text{inf})$ 
// Find g s.t.  $S(i, k) = S(i, k-1) - g(k)$  ( $S(i / p_k, k-1) - S(p_k-1, k-1)$ )
```

```
vector<ll> small, large; // small[i] : S(i) for i = 1
... r^(1/2), large[i] : S(r/i) for i = 1 ... r
^(1/2)
ll get(ll x){ return x <= rt ? small[x] : large[r / x];
}
ll init(ll x){ return x * (x+1) / 2 - 1; }
void lucy(){
    rt = ll(sqrt(r));
    small.resize(rt + 1); large.resize(rt + 1);
    for(int i=1;i<=rt;i++){
        small[i] = init(i); large[i] = init(r / i);
    }
    for(ll k=2;k<=rt;k++){
        if(small[k] == small[k-1]) continue;
        for(ll i = 1; i <= rt && i * k * k <= r; i++){
            large[i] -= k * (get(r / i / k) - small[k-1]);
        }
        for(ll i = rt; i >= k * k; i--){
            small[i] -= k * (small[i/k] - small[k-1]);
        }
    }
}
```

6.10 Quadratic Residue [ac688c]

```
int Jacobi(int a, int m) { // 00fa11
    int s = 1;
    for (; m > 1; ) {
        a %= m;
        if (a == 0) return 0;
        const int r = __builtin_ctz(a);
        if ((r & 1) && ((m + 2) & 4)) s = -s;
        a >>= r;
        if (a & m & 2) s = -s;
        swap(a, m);
    }
    return s;
}
int QuadraticResidue(int a, int p) { // 1a2d08
    if (p == 2) return a & 1;
    const int jc = Jacobi(a, p);
    if (jc == 0) return 0;
    if (jc == -1) return -1;
    int b, d;
    for (; ; ) {
        b = rand() % p;
        d = (11l * b * b + p - a) % p;
        if (Jacobi(d, p) == -1) break;
    }
    int f0 = b, f1 = 1, g0 = 1, g1 = 0, tmp;
    for (int e = (11l + p) >> 1; e; e >>= 1) {
        if (e & 1) {
            tmp = (11l * g0 * f0 + 11l * d * (11l * g1 * f1 %
                p)) % p;
            g1 = (11l * g0 * f1 + 11l * g1 * f0) % p;
            g0 = tmp;
        }
        tmp = (11l * f0 * f0 + 11l * d * (11l * f1 * f1 % p
            )) % p;
        f1 = (21l * f0 * f1) % p;
        f0 = tmp;
    }
    return g0;
}
```

6.11 Simplex [ece29e]

```
struct Simplex { // 0-based
    using T = long double;
    static const int N = 410, M = 30010;
    const T eps = 1e-7;
    int n, m;
    int Left[M], Down[N];
    // Ax <= b, max c^T x. result : v, xi = sol[i]. 1-
    based
    T a[M][N], b[M], c[N], v, sol[N];
    bool eq(T a, T b) {return fabs(a - b) < eps;}
    bool ls(T a, T b) {return a < b && !eq(a, b);}
    void init(int _n, int _m) { // a752a0
        n = _n, m = _m, v = 0;
        for (int i = 0; i < m; ++i) for (int j = 0; j < n;
            ++j) a[i][j] = 0;
```

```
for (int i = 0; i < m; ++i) b[i] = 0;
for (int i = 0; i < n; ++i) c[i] = sol[i] = 0;
}
void pivot(int x, int y) { // 92faa6
    swap(Left[x], Down[y]);
    T k = a[x][y]; a[x][y] = 1;
    vector<int> nz;
    for (int i = 0; i < n; ++i) {
        a[x][i] /= k;
        if (!eq(a[x][i], 0)) nz.push_back(i);
    }
    b[x] /= k;
    for (int i = 0; i < m; ++i) {
        if (i == x || eq(a[i][y], 0)) continue;
        k = a[i][y], a[i][y] = 0;
        b[i] -= k * b[x];
        for (int j : nz) a[i][j] -= k * a[x][j];
    }
    if (eq(c[y], 0)) return;
    k = c[y], c[y] = 0, v += k * b[x];
    for (int i : nz) c[i] -= k * a[x][i];
}
// 0: found solution, 1: no feasible solution, 2:
unbounded
int solve() { // 9f4550
    for (int i = 0; i < n; ++i) Down[i] = i;
    for (int i = 0; i < m; ++i) Left[i] = n + i;
    while (1) {
        int x = -1, y = -1;
        for (int i = 0; i < m; ++i) if (ls(b[i], 0) && (x
            == -1 || b[i] < b[x])) x = i;
        if (x == -1) break;
        for (int i = 0; i < n; ++i) if (ls(a[x][i], 0) &&
            (y == -1 || a[x][i] < a[x][y])) y = i;
        if (y == -1) return 1;
        pivot(x, y);
    }
    while (1) {
        int x = -1, y = -1;
        for (int i = 0; i < n; ++i) if (ls(0, c[i]) && (y
            == -1 || c[i] > c[y])) y = i;
        if (y == -1) break;
        for (int i = 0; i < m; ++i) if (ls(0, a[i][y]) &&
            (x == -1 || b[i] / a[i][y] < b[x] / a[x][y]))
            x = i;
        if (x == -1) return 2;
        pivot(x, y);
    }
    for (int i = 0; i < m; ++i) if (Left[i] < n) sol[
        Left[i]] = b[i];
    return 0;
}
};
```

6.12 Linear Programming Construction

Standard form: maximize $c^T x$ subject to $Ax \leq b$ and $x \geq 0$.
 Dual LP: minimize $b^T y$ subject to $A^T y \geq c$ and $y \geq 0$.
 $\bar{x}$ and $\bar{y}$ are optimal if and only if for all $i \in [1, n]$, either $\bar{x}_i = 0$ or $\sum_{j=1}^m A_{ji} \bar{y}_j = c_i$ holds and for all $i \in [1, m]$ either $\bar{y}_i = 0$ or $\sum_{j=1}^n A_{ij} \bar{x}_j = b_j$ holds.

1. In case of minimization, let $c'_i = -c_i$
2. $\sum_{1 \leq i \leq n} A_{ji} x_i \geq b_j \rightarrow \sum_{1 \leq i \leq n} -A_{ji} x_i \leq -b_j$
3. $\sum_{1 \leq i \leq n} A_{ji} x_i = b_j$
 - $\sum_{1 \leq i \leq n} A_{ji} x_i \leq b_j$
 - $\sum_{1 \leq i \leq n} A_{ji} x_i \geq b_j$
4. If x_i has no lower bound, replace x_i with $x_i - x'_i$

6.13 Estimation

- Number of partitions of n :

n	2	3	4	5	6	7	8	9	20	30	40	50	100
$p(n)$	2	3	5	7	11	15	22	30	627	5604	4e4	2e5	2e8
- Maximum number of divisors:

n	100	1e3	1e6	1e9	1e12	1e15	1e18
$d(i)$	12	32	240	1344	6720	26880	103680
arg	60	840	720720	735134400	963761198400	866421317361600	89761248478661
- $\frac{n}{(2n)}$

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$\frac{n}{(2n)}$	2	6	20	70	252	924	3432	12870	48620	184756	7e5	2e6	1e7	4e7	1.5e8
- Number of ways to partition a set of n labeled elements:

n	2	3	4	5	6	7	8	9	10	11	12	13
B_n	2	5	15	52	203	877	4140	21147	115975	7e5	4e6	3e7

6.14 Theorem

Kirchhoff's Theorem

Denote L be a $n \times n$ matrix as the Laplacian matrix of graph G , where $L_{ii} = d(i)$, $L_{ij} = -c$ where c is the number of edge (i, j) in G .

- The number of undirected spanning in G is $|\det(\tilde{L}_{11})|$.
- The number of directed spanning tree rooted at r in G is $|\det(\tilde{L}_{rr})|$.

Tutte's Matrix

Let D be a $n \times n$ matrix, where $d_{ij} = x_{ij}$ (x_{ij} is chosen uniformly at random) if $i < j$ and $(i, j) \in E$, otherwise $d_{ij} = -d_{ji}$. $\frac{rank(D)}{2}$ is the maximum matching on G .

Cayley's Formula

- Given a degree sequence $d_1, d_2, \dots, d_n$ for each *labeled* vertices, there are

$$\frac{(n-2)!}{(d_1-1)!(d_2-1)! \cdots (d_n-1)!}$$

spanning trees.

- Let $T_{n,k}$ be the number of *labeled* forests on n vertices with k components, such that vertex $1, 2, \dots, k$ belong to different components. Then $T_{n,k} = kn^{n-k-1}$.

Erdős-Gallai Theorem

A sequence of non-negative integers $d_1 \geq d_2 \geq \dots \geq d_n$ can be represented as the degree sequence of a finite simple graph on n vertices if and only if $d_1 + d_2 + \dots + d_n$ is even and

$$\sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min(d_i, k)$$

holds for all $1 \leq k \leq n$.

Burnside's Lemma

Let X be a set and G be a group that acts on X . For $g \in G$, denote by X^g the elements fixed by g :

$$X^g = \{x \in X \mid gx = x\}$$

Then

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|.$$

Gale-Ryser theorem

A pair of sequences of nonnegative integers $a_1 \geq \dots \geq a_n$ and $b_1, \dots, b_n$ is bigraphic if and only if $\sum_{i=1}^n a_i = \sum_{i=1}^n b_i$ and $\sum_{i=1}^k a_i \leq \sum_{i=1}^n \min(b_i, k)$ holds for every $1 \leq k \leq n$.

Fulkerson-Chen-Anstee theorem

A sequence $(a_1, b_1), \dots, (a_n, b_n)$ of nonnegative integer pairs with $a_1 \geq \dots \geq a_n$ is digraphic if and only if $\sum_{i=1}^n a_i = \sum_{i=1}^n b_i$ and $\sum_{i=1}^k a_i \leq \sum_{i=1}^n \min(b_i, k)$ holds for every $1 \leq k \leq n$.

$$\sum_{i=1}^k \min(b_i, k-1) + \sum_{i=k+1}^n \min(b_i, k) \text{ holds for every } 1 \leq k \leq n.$$

Möbius inversion formula

- $f(n) = \sum_{d|n} g(d) \Leftrightarrow g(n) = \sum_{d|n} \mu(d) f(\frac{n}{d})$
- $f(n) = \sum_{n|d} g(d) \Leftrightarrow g(n) = \sum_{n|d} \mu(\frac{d}{n}) f(d)$

Spherical cap

- A portion of a sphere cut off by a plane.
- r : sphere radius, a : radius of the base of the cap, h : height of the cap, θ : $\arcsin(a/r)$.
- Volume $= \pi h^2(3r-h)/3 = \pi h(3a^2+h^2)/6 = \pi r^3(2+\cos\theta)(1-\cos\theta)^2/3$.
- Area $= 2\pi rh = \pi(a^2+h^2) = 2\pi r^2(1-\cos\theta)$.

Chinese Remainder Theorem

- $x \equiv a_i \pmod{m_i}$
- $M = \prod m_i, M_i = M/m_i$
- $t_i M_i \equiv 1 \pmod{m_i}$
- $x = \sum a_i t_i M_i \pmod{M}$

- Green Hackenbush: Run BCC(bridge), each BCC has Grundy value $g_v = |E| \bmod 2$ (the Grundy value remains unchanged if we identify a circuit as a node). Run DFS with root *ground* and $g_v \leftarrow g_v \oplus (g_h + 1)$ for vertex v and its child h .

6.15 General Purpose Numbers

Bernoulli numbers

$$B_0 = 1, B_1^\pm = \pm \frac{1}{2}, B_2 = \frac{1}{6}, B_3 = 0$$

$$\sum_{j=0}^m \binom{m+1}{j} B_j = 0, \text{ EGF is } B(x) = \frac{x}{e^x - 1} = \sum_{n=0}^{\infty} B_n \frac{x^n}{n!}.$$

$$S_m(n) = \sum_{k=1}^n k^m = \frac{1}{m+1} \sum_{k=0}^m \binom{m+1}{k} B_k^+ n^{m+1-k}$$

- Stirling numbers of the second kind Partitions of n distinct elements into exactly k groups.

$$S(n, k) = S(n-1, k-1) + kS(n-1, k), S(n, 1) = S(n, n) = 1$$

$$S(n, k) = \frac{1}{k!} \sum_{i=0}^k (-1)^{k-i} \binom{k}{i} i^n$$

$$x^n = \sum_{i=0}^n S(n, i) (x)_i$$

Pentagonal number theorem

$$\prod_{n=1}^{\infty} (1 - x^n) = 1 + \sum_{k=1}^{\infty} (-1)^k \left(x^{k(3k+1)/2} + x^{k(3k-1)/2} \right)$$

Catalan numbers

$$C_n^{(k)} = \frac{1}{(k-1)n+1} \binom{kn}{n}$$

$$C^{(k)}(x) = 1 + x[C^{(k)}(x)]^k$$

Eulerian numbers

Number of permutations $\pi \in S_n$ in which exactly k elements are greater than the previous element. k : j 's s.t. $\pi(j) > \pi(j+1)$, $k+1$: j 's s.t. $\pi(j) \geq j$, k : j 's s.t. $\pi(j) > j$.

$$E(n, k) = (n-k)E(n-1, k-1) + (k+1)E(n-1, k)$$

$$E(n, 0) = E(n, n-1) = 1$$

$$E(n, k) = \sum_{j=0}^k (-1)^j \binom{n+1}{j} (k+1-j)^n$$

6.16 Integral

$\frac{d}{dx}$	$\sec^2 x$	$-\csc^2 x$	$\tan x \sec x$	$-\cot x \csc x$
$f(x)$	$\tan x$	$\cot x$	$\sec x$	$\csc x$
$\int$	$\ln \sec x $	$-\ln \csc x $	$\ln \tan x + \sec x $	$-\ln \cot x + \csc x $
$\frac{d}{dx}$	$\frac{1}{\sqrt{1-x^2}}$	$\frac{-1}{\sqrt{1-x^2}}$	$\frac{1}{x^2+1}$	$\sec^3 x$
$f(x)$	$\arcsin x$	$\arccos x$	$\arctan x$	$\frac{1}{2} (\sec x \tan x + \ln \sec x + \tan x)$

$$\int_0^{\frac{\pi}{2}} \sin^{2k} x \, dx = \frac{\pi}{2} \cdot \frac{1 \cdot 3 \cdots (2k-1)}{2 \cdot 4 \cdots (2k)} \quad \int_0^{\frac{\pi}{2}} \sin^{2k+1} x \, dx = \frac{2 \cdot 4 \cdots (2k)}{3 \cdot 5 \cdots (2k+1)}$$

$$\iint f(x, y) \, dx \, dy = \iint f(g(u, v), h(u, v)) \left| \frac{\partial x}{\partial u} \frac{\partial x}{\partial v} \right| \, du \, dv$$

$$\text{polar: } \iint r \cdot f(r \cos \theta, r \sin \theta) \, dr \, d\theta$$

$$\text{spherical: } \iiint \rho^2 \sin \phi \cdot f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \theta) \, d\phi \, d\theta \, d\rho$$

$$\text{Green's theorem: } \iint \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \, dx \, dy = \oint (P \, dx + Q \, dy)$$

Area: $A = \oint x \, dy = -\oint y \, dx$ (counter-clockwise)

Circular sector: $(x, y) = (x_0 + r \cos \theta, y_0 + r \sin \theta)$,

$$\int_{\alpha}^{\beta} x \, dy - y \, dx = r(r\theta + x_0 \sin \theta - y_0 \cos \theta) \Big|_{\alpha}^{\beta}$$

6.17 Floor Sum

$$m = \lfloor \frac{a+b}{c} \rfloor$$

$$\text{Time complexity: } O(\log n)$$

$$f(a, b, c, n) = \sum_{i=0}^n \lfloor \frac{ai+b}{c} \rfloor$$

$$= \begin{cases} \lfloor \frac{a}{c} \rfloor \cdot \frac{n(n+1)}{2} + \lfloor \frac{b}{c} \rfloor \cdot (n+1) \\ + f(a \bmod c, b \bmod c, c, n), & a \geq c \vee b \geq c \\ 0, & n < 0 \vee a = 0 \\ nm - f(c, c-b-1, a, m-1), & \text{otherwise} \end{cases}$$

$$g(a, b, c, n) = \sum_{i=0}^n i \lfloor \frac{ai+b}{c} \rfloor$$

$$= \begin{cases} \lfloor \frac{a}{c} \rfloor \cdot \frac{n(n+1)(2n+1)}{6} + \lfloor \frac{b}{c} \rfloor \cdot \frac{n(n+1)}{2} \\ + g(a \bmod c, b \bmod c, c, n), & a \geq c \vee b \geq c \\ 0, & n < 0 \vee a = 0 \\ \frac{1}{2} \cdot (n(n+1)m - f(c, c-b-1, a, m-1) \\ - h(c, c-b-1, a, m-1)), & \text{otherwise} \end{cases}$$

$$h(a, b, c, n) = \sum_{i=0}^n \left\lfloor \frac{ai+b}{c} \right\rfloor^2$$

$$= \begin{cases} \left\lfloor \frac{a}{c} \right\rfloor^2 \cdot \frac{n(n+1)(2n+1)}{6} + \left\lfloor \frac{b}{c} \right\rfloor^2 \cdot (n+1) \\ + \left\lfloor \frac{a}{c} \right\rfloor \cdot \left\lfloor \frac{b}{c} \right\rfloor \cdot n(n+1) \\ + h(a \bmod c, b \bmod c, c, n) \\ + 2 \left\lfloor \frac{a}{c} \right\rfloor \cdot g(a \bmod c, b \bmod c, c, n) \\ + 2 \left\lfloor \frac{b}{c} \right\rfloor \cdot f(a \bmod c, b \bmod c, c, n), & a \geq c \vee b \geq c \\ 0, & n < 0 \vee a = 0 \\ nm(m+1) - 2g(c, c-b-1, a, m-1) \\ - 2f(c, c-b-1, a, m-1) - f(a, b, c, n), & \text{otherwise} \end{cases}$$

7 Polynomial

7.1 FFT [43cc28]

```
const double pi=acos(-1);
typedef complex<double> cp;
const int N=(1<<17);
struct FFT
{
    // n has to be same as a.size()
    int n, rev[N];
    cp omega[N], iomega[N];
    void init(int _n) {
        n=_n;
        for(int i=0; i<n; i++) {
            omega[i]=cp(cos(2*pi/n*i), sin(2*pi/n*i));
            iomega[i]=conj(omega[i]);
        }
        int k=log2(n);
        for(int i=0; i<n; i++) {
            rev[i]=0;
            for(int j=0; j<k; j++) if(i&(1<<j))
                rev[i]|=(1<<(k-j-1));
        }
    }
    void tran(vector<cp> &a, cp* xomega)
    {
        for(int i=0; i<n; i++) if(i<rev[i])
            swap(a[i], a[rev[i]]);
        for(int len=2; len<=n; len<=1) {
            int mid=len>>1, r=n/len;
            for(int j=0; j<n; j+=len) {
                for(int i=0; i<mid; i++) {
                    cp t=xomega[r*i]*a[j+mid+i];
                    a[j+mid+i]=a[j+i]-t;
                    a[j+i]+=t;
                }
            }
        }
    }
    void fft(vector<cp> &a) {tran(a, omega);}
    void ifft(vector<cp> &a) {
        tran(a, iomega);
        for(int i=0; i<n; i++) a[i]/=n;
    }
};
```

7.2 NTT [39f8b1]

```
//(2^16)+1, 65537, 3
//7*17*(2^23)+1, 998244353, 3
//1255*(2^20)+1, 1315962881, 3
//51*(2^25)+1, 1711276033, 29
// only works when sz(A) + sz(B) - 1 <= MAXN
// implement add, po, mul, inv first
template<int N, int RT>
struct NTT{
    int w[N];
    NTT() {
        int dw = po(RT, (mod-1) / N);
        w[0] = 1;
        for(int i=1; i<N; i++) w[i] = mul(w[i-1], dw);
    }
    void bitrev(vector<int>& a, int n){
        int i=0;
        for(int j=1; j<n-1; j++) {
            for(int k=n>>1; (i^=k) < k; k>>=1);
            if(j<i) swap(a[i], a[j]);
        }
    }
```

```
}
void operator()(vector<int>& a, int n, bool ix = false)
{
    bitrev(a, n);
    for(int L=2; L<=n; L<=1){
        int dx = N/L, dl = L>>1;
        for(int i=0; i<n; i+=L){
            for(int j=i, x=0; j<i+dl; j++, x+=dx){
                int tmp = mul(a[j+dl], w[x]);
                a[j+dl] = a[j];
                add(a[j+dl], -tmp);
                add(a[j], tmp);
            }
        }
    }
    if(ix){
        reverse(a.begin()+1, a.begin()+n);
        int invn = inv(n);
        for(int i=0; i<n; i++) a[i] = mul(a[i], invn);
    }
};
```

7.3 FWT [d4ce4e]

```
/* (op, invop): (oyx: op from y to x)
or: (1, 0, 1, 1), (1, 0, -1, 1)
and: (1, 1, 0, 1), (1, -1, 0, 1)
xor: (1, 1, 1, -1), (1/2, 1/2, 1/2, -1/2) */
void fwt(vector<int>& a, int n, int oxx, int oyx, int
        oxy, int oyy){ // 2^20, 0.5s
    for(int L=2; L<=n; L<=1)
        for(int i=0; i<n; i+=L)
            for(int j=i; j<i+(L>>1); j++){
#define x a[j]
#define y a[j+(L>>1)]
                tie(x, y) = pii(mul(x, oxx)+mul(y, oyx), mul(x,
                    oxy)+mul(y, oyy));
                add(x, 0); add(y, 0);
#undef x
#undef y
            }
    }
    vector<int> subset_conv(vector<int> a, vector<int> b,
        int L){
        int n = 1<<L; // 2^20, 5.5s, c-style arrays -> 2.1s
        vector<vector<int>> f(L+1, vector<int>(n)), g=f, h=f;
        vector<int> ct(n), c(n);
        for(int i=1; i<n; i++) ct[i] = ct[i&(i-1)] + 1;
        for(int i=0; i<n; i++) { f[ct[i]][i] = a[i]; g[ct[i]][i]
            = b[i]; }
        for(int i=0; i<=L; i++) { fwt(f[i], n, 1, 0, 1, 1); fwt
            (g[i], n, 1, 0, 1, 1); }
        for(int i=0; i<=L; i++)
            for(int j=0; j<=i; j++)
                for(int x=0; x<n; x++)
                    add(h[i][x], mul(f[j][x], g[i-j][x]));
        for(int i=0; i<=L; i++) fwt(h[i], n, 1, 0, -1, 1);
        for(int i=0; i<n; i++) c[i] = h[ct[i]][i];
        return c;
    }
```

7.4 Polynomial [ede4d0]

```
// may need fac ivf
// == ab8066 ==
#define fi(s, n) for (int i = (int)(s); i < (int)(n); i++)
#define neg(x) (x ? mod - x : 0)
#define V (*this)
template<int MAXN, int RT> // MAXN = 2^k
struct Poly : vector<int> { // coefficients in [0, P)
    using vector<int>::vector;
    static inline NTT<MAXN, RT> ntt;
    int n() const { return (int)size(); } // n() >= 1
    Poly(const Poly &p, int m) : vector<int>(m) { copy_n(
        p.data(), min(p.n(), m), data()); }
    Poly &irev() { return reverse(data(), data() + n()),
        V; }
    Poly &isz(int m) { return resize(m), V; }
    // == e33faa ==
    Poly &iadd(const Poly &rhs) { // 1c6277
```

```

    fi(0, n()) add(V[i], rhs[i]);
    return V; // need n() == rhs.n()
}
Poly &imul(int k) { // 7e5b36
    fi(0, n()) V[i] = mul(V[i], k);
    return V;
}
Poly &mul_xk(int m) { // 19862a
    if(m<0) erase(begin(), begin()+(-m));
    else if(m>0) insert(begin(), m, 0);
    return V;
}
Poly Mul(const Poly &rhs) const { // ecd03e
    int m = 1;
    while (m < n() + rhs.n() - 1) m <= 1;
    assert(m <= MAXN);
    Poly X(V, m), Y(rhs, m);
    ntt(X, m); ntt(Y, m);
    fi(0, m) X[i] = mul(X[i], Y[i]);
    ntt(X, m, true);
    return X.isz(n() + rhs.n() - 1);
}
Poly Inv() const { // 77f977
    if (n() == 1) return {inv(V[0])};
    int m = 1; // need V[0] != 0, 2*sz<=MAXN
    while (m < n() * 2) m <= 1;
    assert(m <= MAXN);
    Poly Xi = Poly(V, (n() + 1) / 2).Inv().isz(m);
    Poly Y(V, m);
    ntt(Xi, m); ntt(Y, m);
    fi(0, m) {
        Xi[i] = mul(Xi[i], (2 - mul(Xi[i], Y[i])));
        add(Xi[i], mod);
    }
    ntt(Xi, m, true);
    return Xi.isz(n());
}
// == 095701 ==
Poly &shift_inplace(const int &c) { // need fac[],
    ivf[]
    int n = V.n(); // 2 * sz <= MAXN
    for (int i = 0; i < n; i++) V[i] = mul(V[i], fac[i]);
    Poly g(n);
    int cp = 1;
    for (int i = 0; i < n; i++){
        g[i] = mul(cp, ivf[i]);
        cp = mul(cp, c);
    }
    V = V.irev().Mul(g).isz(n).irev();
    for (int i = 0; i < n; i++) V[i] = mul(V[i], ivf[i]);
    return V;
}
Poly Shift(const int &c) const { return Poly(V).
    shift_inplace(c); }
// == 0bd61d ==
Poly Dx() const {
    Poly ret(n() - 1);
    fi(0, ret.n()) ret[i] = mul(i+1, V[i+1]);
    return ret.isz(max(1, ret.n()));
}
Poly Sx() const {
    Poly ret(n() + 1);
    fi(0, n()) ret[i + 1] = mul(inv(i+1), V[i]);
    return ret;
}
// == 10e23d ==
Poly Ln() const { // V[0] == 1, 2*sz<=MAXN
    return Dx().Mul(Inv()).Sx().isz(n());
}
Poly Exp() const { // V[0] == 0, 2*sz<=MAXN
    if (n() == 1) return {1};
    Poly X = Poly(V, (n() + 1) / 2).Exp().isz(n());
    Poly Y = X.Ln();
    Y[0] = mod - 1;
    fi(0, n()){
        Y[i] = V[i] - Y[i];
        add(Y[i], mod);
    }
    return X.Mul(Y).isz(n());
}
// == eaf14c ==
// M := P(P - 1). If k >= M, k := k % M + M.
Poly Pow(int k) const { // 2*sz<=MAXN
    int nz = 0;
    while (nz < n() && !V[nz]) nz++;
    if (nz * min(k, (11)n()) >= n()) return Poly(n());
    if (!k) return Poly(Poly{1}, n());
    Poly X(data() + nz, data() + nz + n() - nz * k);
    const int c = po(X[0], k % (mod - 1));
    return X.Ln().imul(k % mod).Exp().imul(c).irev().
        isz(n()).irev();
}
// == cdf741 ==
Poly _tmul(int nn, const Poly &rhs) const {
    Poly Y = Mul(rhs).isz(n() + nn - 1);
    return Poly(Y.data() + n() - 1, Y.data() + Y.n());
}
vector<int> _eval(const vector<int> &x, const vector<
    Poly> &up) const { // 82d6be
    const int m = (int)x.size();
    if (!m) return {};
    vector<Poly> down(m * 2);
    down[1] = Poly(up[1]).irev().isz(n()).Inv().irev().
        _tmul(m, V);
    fi(2, m * 2) down[i] = up[i ^ 1]._tmul(up[i].n() -
        1, down[i / 2]);
    vector<int> y(m);
    fi(0, m) y[i] = down[m + i][0];
    return y;
}
static vector<Poly> _tree1(const vector<int> &x) { //
    2a0b6b
    const int m = (int)x.size();
    vector<Poly> up(m * 2);
    fi(0, m) up[m + i] = {neg(x[i]), 1};
    for (int i = m - 1; i > 0; --i) up[i] = up[i * 2].
        Mul(up[i * 2 + 1]);
    return up;
}
vector<int> Eval(const vector<int> &x) const {
    auto up = _tree1(x); // 2^17, 1.8s
    return _eval(x, up);
}
// == f9ecd4 ==
static Poly Interpolate(const vector<int> &x, const
    vector<int> &y) { // 8f2a08
    const int m = (int)x.size(); // 2^17, 2.3s
    vector<Poly> up = _tree1(x), down(m * 2);
    vector<int> z = up[1].Dx()._eval(x, up);
    fi(0, m) z[i] = mul(y[i], inv(z[i]));
    fi(0, m) down[m + i] = {z[i]};
    for (int i = m - 1; i > 0; --i)
        down[i] = down[i * 2].Mul(up[i * 2 + 1]).iadd(
            down[i * 2 + 1].Mul(up[i * 2]));
    return down[1];
}
pair<Poly, Poly> DivMod(const Poly &rhs) const { //
    a75170
    if (n() < rhs.n()) return {{0}, V}; // 5e5, 0.9s
    const int m = n() - rhs.n() + 1;
    Poly X(rhs); // (rhs).back() != 0
    X.irev().isz(m);
    Poly Y(V); Y.irev().isz(m);
    Poly Q = Y.Mul(X.Inv()).isz(m).irev();
    X = rhs.Mul(Q), Y = V;
    fi(0, n()) add(Y[i], -X[i]);
    return {Q, Y.isz(max(1, rhs.n() - 1))};
}
// == 7cd4c4 ==
Poly _Sqrt() const { // Jacobi(V[0], P) = 1
    if (n() == 1) return {QuadraticResidue(V[0], mod)};
    Poly X = Poly(V, (n() + 1) / 2)._Sqrt().isz(n());
    return X.iadd(Mul(X.Inv()).isz(n())).imul(mod / 2 +
        1);
}
Poly Sqrt() const { // 288427
    Poly a; // 2 * sz <= MAXN
    bool has = 0;
    for (int i = 0; i < n(); i++) {
        if (V[i]) has = 1;
        if (has) a.push_back(V[i]);
    }
}

```

```

    if (!has) return V;
    if ((n() + a.n()) % 2 || Jacobi(a[0], mod) != 1) {
        return Poly();
    }
    a = a.isz((n() + a.n()) / 2)._Sqrt();
    int sz = a.n();
    a.isz(n());
    rotate(a.begin(), a.begin() + sz, a.end());
    return a;
}
// == 0a54cf ==
Poly Shift_samples(int c, int m){
    // V = \sum f(n) x^n, return f(c), ..., f(c+m-1);
    // 2^19, 2s
    Poly A=V;
    Poly Q(n()+1), S(n());
    int nw=1;
    fi(0, n()+1) { Q[i] = mul(1-2*(i&1), nw); nw = mul(
        nw, n()-i, inv(i+1)); }
    nw=1;
    fi(0, n()) { S[i] = mul(1-2*(i&1), nw); nw = mul(nw,
        c-i, inv(i+1)); }
    S=S.Shift(1);
    fi(0, n()) if(i&1) { S[i] = mul(S[i], -1); add(S[i],
        mod); }
    Poly C=Mul(Q).mul_xk(-n());
    auto tmp=Q.isz(m).Inv();
    C=C.imul(-1).Mul(tmp).isz(m).mul_xk(n());
    A=A.isz(n()+m).iadd(C).irev().isz(n()+m);
    return A.Mul(S).mul_xk(-n()+1).isz(m+1).irev();
}
// == 224b20 ==
Poly power_projection(Poly wt, int m) { // 857237
    assert(n() == wt.n()); // 4*sz <= MAXN
    if (!n()) return Poly(m + 1, 0);
    if (V[0] != 0) {
        int c = V[0];
        V[0] = 0;
        Poly A = V.power_projection(wt, m);
        fi(0, m + 1) A[i] = mul(A[i], fac[i]);
        Poly B(m + 1);
        int pow = 1;
        fi(0, m + 1) { B[i] = mul(pow, ivf[i]); pow = mul(
            pow, c); } // inv. of fac
        A = A.Mul(B).isz(m + 1);
        fi(0, m + 1) A[i] = mul(A[i], fac[i]);
        return A;
    }
    int n = 1;
    while (n < V.n()) n *= 2;
    isz(n), wt.isz(n).irev();
    int k = 1;
    Poly p(wt, 2 * n), q(V, 2 * n);
    q.imul(mod - 1);
    while (n > 1) {
        Poly r(2 * n * k);
        fi(0, 2 * n * k) r[i] = (i % 2 == 0 ? q[i] : neg(
            q[i]));
        Poly pq = p.Mul(r).isz(4 * n * k);
        Poly qq = q.Mul(r).isz(4 * n * k);
        fi(0, 2 * n * k) {
            add(pq[2 * n * k + i], p[i]);
            add(qq[2 * n * k + i], (q[i] + r[i]) % mod);
        }
        fill(p.begin(), p.end(), 0);
        fill(q.begin(), q.end(), 0);
        for(int j = 0; j < 2 * k; j++) fi(0, n / 2) {
            p[n * j + i] = pq[(2 * n) * j + (2 * i + 1)];
            q[n * j + i] = qq[(2 * n) * j + (2 * i + 0)];
        }
        n /= 2; k *= 2;
    }
    Poly ans(k);
    fi(0, k) ans[i] = p[2 * i];
    return ans.irev().isz(m + 1);
}
Poly FPSinv() { // 49e5ad
    const int n = V.n() - 1; // 2^17, 4s
    if (n == -1) return {};
    assert(V[0] == 0);
    if (n == 0) return V;
    assert(V[1] != 0);

```

```

    int c = V[1], ic = inv(c);
    imul(ic);
    Poly wt(n + 1);
    wt[n] = 1;
    Poly A = V.power_projection(wt, n);
    Poly g(n);
    fi(1, n + 1) g[n - i] = mul(n, A[i], inv(i));
    g = g.Pow(neg(inv(n)));
    g.insert(g.begin(), 0);
    int pow = 1;
    fi(0, g.n()) { g[i] = mul(g[i], pow); pow = mul(pow,
        ic); }
    return g;
}
// == 71ee33 ==
Poly TMul(const Poly &rhs) const { // this[i] - rhs[j]
    j = k;
    return Poly(V.irev().Mul(rhs).isz(n()).irev());
}
Poly comp_rec(int n, int k, Poly Q) { // 06ca07
    if (n == 1) {
        Poly p(2 * k);
        irev();
        fi(0, k) p[2 * i] = V[i];
        return p;
    }
    Poly R(2 * n * k);
    fi(0, 2 * n * k) R[i] = (i % 2 == 0 ? Q[i] : neg(Q[
        i]));
    Poly QQ = Q.Mul(R).isz(4 * n * k);
    fi(0, 2 * n * k)
        add(QQ[2 * n * k + i], (Q[i] + R[i]) % mod);
    Poly nxt_Q(2 * n * k);
    for(int j = 0; j < 2 * k; j++) fi(0, n / 2)
        nxt_Q[n * j + i] = QQ[(2 * n) * j + (2 * i + 0)];
    Poly nxt_p = comp_rec(n / 2, k * 2, nxt_Q);
    Poly pq(4 * n * k);
    for(int j = 0; j < 2 * k; j++) fi(0, n / 2)
        add(pq[(2 * n) * j + (2 * i + 1)], nxt_p[n * j +
            i]);
    Poly p(2 * n * k);
    fi(0, 2 * n * k) add(p[i], pq[2 * n * k + i]);
    pq.pop_back();
    Poly x = pq.TMul(R);
    fi(0, 2 * n * k) add(p[i], x[i]);
    return p;
}
Poly FPScomp(Poly g) { // solves V(g(x))
    int sz = 1; // 2^17, 5s
    while(sz < n() || sz < g.n()) sz <= 1;
    return isz(sz), comp_rec(sz, 1, g.imul(mod-1).isz(2
        * sz)).isz(sz).irev();
}
// == e42b36 ==
};
#undef V
#undef neg
#undef fi
using Poly_t = Poly<1 << 21, 3>;

```

7.5 BM-BM [362501]

```

// out[i] = me[0]*out[i-1] + me[1]*out[i-2]+...
vector<int> BerlekampMassey(const vector<int> &output)
{ // 2e7709
    vector<int> d(output.size() + 1), me, he;
    for (int f = 0, i = 1; i <= sz(output); ++i) {
        for (int j = 0; j < sz(me); ++j)
            add(d[i], mul(output[i - j - 2], me[j]));
        add(d[i], -output[i-1]);
        if (d[i] == 0) continue;
        if (me.empty()) {
            me.resize(f = i);
            continue;
        }
        vector<int> o(i - f - 1);
        int k = mul(d[i], -inv(d[f]));
        o.push_back(-k);
        for (auto x : he) o.push_back(mul(x, k));
        if (o.size() < me.size()) o.resize(me.size());
        for (size_t j = 0; j < me.size(); ++j) add(o[j], me
            [j]);
    }
}

```

```

    if (i-f+he.size() >= me.size()) he = me, f = i;
    me = o;
}
for(auto &h: me) add(h, mod);
return me;
}

// Finds the k-th coefficient of P / Q in O(d log d log k)
// size of NTT has to > 2 * d
NTT<1<<19, 3> ntt;
int BostanMori(vector<int> P, vector<int> Q, long long k) { // 1bee33
    int d = max((int)P.size(), (int)Q.size() - 1);
    vector M = {P, Q};
    M[0].resize(d, 0);
    M[1].resize(d + 1, 0);
    int sz = (2 * d + 1 == 1 ? 2 : (1 << (1 + __lg(2 * d) + 1)));
    vector<int> Qn(sz);
    vector N(2, vector<int>(sz));
    while(k) {
        fill(iter(Qn), 0);
        for(int i = 0; i < d + 1; i++){
            Qn[i] = M[1][i] * ((i & 1) ? -1 : 1);
            add(Qn[i], mod);
        }
        ntt(Qn, sz, false);
        int t[2] = {(int)k & 1, 0};
        for(int i = 0; i < 2; i++){
            fill(iter(N[i]), 0);
            copy(iter(M[i]), N[i].begin());
            ntt(N[i], sz, false);
            for(int j = 0; j < sz; j++) N[i][j] = mul(N[i][j], Qn[j]);
            ntt(N[i], sz, true);
            for(int j = t[i]; j < 2 * sz(M[i]); j += 2){
                M[i][j >> 1] = N[i][j];
            }
        }
        k >>= 1;
    }
    return mul(M[0][0], inv(M[1][0]));
}

// a_n = \sum_{j=1}^d a_j a_{n-j}, c_0 = 0
int LinearRecursion(vector<int> a, vector<int> c, ll k)
{ // 3b1227
    int d = sz(a), sz = 1; // 1e5, 5s
    while(sz <= 2*d) sz <<= 1;
    c[0] = mod - 1;
    for(auto &i : c) i = i ? mod - i : 0;
    auto A = a; A.resize(sz);
    auto C = c; C.resize(sz);
    ntt(A, sz, false), ntt(C, sz, false);
    for(int i = 0; i < sz; i++) A[i] = mul(A[i], C[i]);
    ntt(A, sz, true);
    A.resize(d);
    return BostanMori(A, c, k);
}

```

7.6 Generating Functions

- Ordinary Generating Function $A(x) = \sum_{i \geq 0} a_i x^i$

$$\begin{aligned}
 & A(rx) \Rightarrow r^n a_n \\
 & A(x) + B(x) \Rightarrow a_n + b_n \\
 & A(x)B(x) \Rightarrow \sum_{i=0}^n a_i b_{n-i} \\
 & A(x)^k \Rightarrow \sum_{i_1+i_2+\dots+i_k=n} a_{i_1} a_{i_2} \dots a_{i_k} \\
 & xA(x)' \Rightarrow na_n \\
 & \frac{A(x)}{1-x} \Rightarrow \sum_{i=0}^n a_i
 \end{aligned}$$

- Exponential Generating Function $A(x) = \sum_{i \geq 0} \frac{a_i}{i!} x^i$

$$\begin{aligned}
 & A(x) + B(x) \Rightarrow a_n + b_n \\
 & A^{(k)}(x) \Rightarrow a_{n+k} \\
 & A(x)B(x) \Rightarrow \sum_{i=0}^n \binom{n}{i} a_i b_{n-i} \\
 & A(x)^k \Rightarrow \sum_{i_1+i_2+\dots+i_k=n} \binom{n}{i_1, i_2, \dots, i_k} a_{i_1} a_{i_2} \dots a_{i_k} \\
 & xA(x) \Rightarrow na_n
 \end{aligned}$$

- Special Generating Function

$$\begin{aligned}
 & (1+x)^n = \sum_{i \geq 0} \binom{n}{i} x^i \\
 & \frac{1}{(1-x)^n} = \sum_{i \geq 0} \binom{n+i-1}{n-1} x^i
 \end{aligned}$$

- Catalan number: $f(x) = \frac{1-\sqrt{1-4x}}{2x}$

- Suppose $A(x)$ is the EGF of a labelled component. Then $B(x) = e^{A(x)}$ is the EGF of the same thing but with any number of components.

- Lagrange's Inversion Formula

$F(G(x)) = G(F(x)) = x$, then $F(0) = G(0) = 0$ and $F'(0), G'(0) \neq 0$. Let H be any FPS, then

$$n[x^n]F(x) = [x^{n-1}]H'(x) \left(\frac{x}{G(x)} \right)^n.$$

- Newton's Method

Suppose FPS P satisfies $F(x, P(x)) = 0$ where $F(x, y)$ is some polynomial, then

$$P_{2n}(x) = P_n(x) - F(x, P_n(x)) / \frac{\partial F}{\partial y}(x, P_n(x)).$$

8 Geometry

8.1 Basic [d4b33e]

```

// == d2c7dd ==
struct PT {
    ll x, y;
    auto operator<=>(const PT&) const = default;
};
PT operator+(PT p1, PT p2)
{ return PT{p1.x + p2.x, p1.y + p2.y}; }
PT operator-(PT p1, PT p2)
{ return PT{p1.x - p2.x, p1.y - p2.y}; }
ll operator*(PT p1, PT p2)
{ return p1.x * p2.x + p1.y * p2.y; }
ll operator^(PT p1, PT p2)
{ return p1.x * p2.y - p1.y * p2.x; }
int sign(ll a) { return a == 0 ? 0 : a > 0 ? 1 : -1; }
int ori(PT a, PT b, PT c) { // is C to the left of A->B
    return sign((b-a)^(c-a)); }
bool btw(PT a, PT b, PT c) { // is C between AB
    return !ori(a, b, c) && sign((a-c)*(b-c)) <= 0; }
PT ccw90(PT p) { return PT(-p.y, p.x); }
// == f61f45 ==
// change PT::x, y to FP type, replace sign()
// needs PT+-PT, PT*/scalar
const double EPS = 1e-9;
int sign(double a) { return abs(a)<EPS?0:a>0?1:-1; }
double abs2(PT p){ return p * p; }
double abs(PT p){ return sqrt(p * p); }
PT proj(PT a, PT b, PT c){ // C projected to AB
    return a+(b-a)*((c-a)*(b-a)/abs2(b-a)); }
double dist(PT a,PT b,PT c){ // distance from C to AB
    return abs((c-a)^(b-a))/abs(b-a); }
// struct Line{ PT a, b; };
// struct Cir{ PT o; double r; };

```

8.2 Convex Hull [903417]

```

// (int) requires sz(pt) >= 2 & distinct points
// points on edges are excluded
vector<PT> ConvexHull(vector<PT> pt) {
    int n = sz(pt);
    sort(iter(pt));
    vector<PT> ans{pt[0]};
    For(t, 0, 1) {
        int m = sz(ans);
        For(i, 1, n - 1) {
            while (sz(ans) > m && ori(ans[sz(ans) - 2],
                ans.back(), pt[i]) <= 0) ans.pop_back();
            ans.push_back(pt[i]);
        }
        reverse(iter(pt));
    }
    if (sz(ans) > 1) ans.pop_back();
    return ans;
}

```

8.3 Dynamic Convex Hull [95890b]

```
// (int)
struct DynamicConvexHull {
    struct UpCmp {
        bool operator()(PT a, PT b) const {
            return a < b;
        }
    };
    struct DownCmp {
        bool operator()(PT a, PT b) const {
            return a > b;
        }
    };
    template <typename T>
    struct Hull {
        set<PT,T> hull;
        bool chk(PT i, PT j, PT k) {
            return ((k-i)^(j-i)) > 0;
        }
        void insert(PT x) {
            if(inside(x)) return;
            hull.insert(x);
            auto it=hull.lower_bound(x);
            if(next(it)!=hull.end()) {
                for (auto it2 = next(it);
                    next(it2) != hull.end(); ++it2) {
                    if(chk(x,*it2,*next(it2))) break;
                    hull.erase(it2);
                    it2=hull.lower_bound(x);
                }
            }
            it=hull.lower_bound(x);
            if(it!=hull.begin()) {
                for (auto it2 = prev(it);
                    it2 != hull.begin(); --it2) {
                    if(chk(*prev(it2),*it2,x)) break;
                    hull.erase(it2);
                    it2=hull.lower_bound(x);
                    if(it2==hull.begin()) break;
                }
            }
        }
        bool inside(PT x) {
            if (hull.lower_bound(x) != hull.end() &&
                *hull.lower_bound(x) == x)
                return true;
            auto it=hull.lower_bound(x);
            bool ans=false;
            if(it!=hull.begin()&&it!=hull.end()) {
                ans=!chk(*prev(it),x,*it);
            }
            return ans;
        }
    };
    Hull<UpCmp> up;
    Hull<DownCmp> down;
    void insert(PT x){up.insert(x),down.insert(x);}
    bool inside(PT x) {
        return up.inside(x) && down.inside(x);
    }
};
```

8.4 Point In Convex Hull [9a1f2c]

```
// (int) C must be a convex polygon in cyclic order
bool PointInConvex(const vector<PT> &C, PT p,
    bool strict = true) {
    int a = 1, b = int(C.size()) - 1, r = !strict;
    if (C.size() == 0) return false;
    if (C.size() < 3) return r && btw(C[0], C.back(), p);
    if (ori(C[0], C[a], C[b]) > 0) swap(a, b);
    if (ori(C[0], C[a], p) >= r ||
        ori(C[0], C[b], p) <= -r) return false;
    while (abs(a - b) > 1) {
        int c = (a + b) / 2;
        (ori(C[0], C[c], p) > 0 ? b : a) = c;
    }
    return ori(C[a], C[b], p) < r;
}
```

8.5 Point In Circle [9d9594]

```
// (int) p1, p2, p3 must be in CCW order
```

```
// test if p4 strictly in circumscribed circle of p1p2p3
bool InCir(PT p1, PT p2, PT p3, PT p4) {
    p1 = p1 - p4; p2 = p2 - p4; p3 = p3 - p4;
    ll l1 = p1 * p1, l2 = p2 * p2, l3 = p3 * p3;
    i128 det = (i128)l1 * (p2 ^ p3)
        + (i128)l2 * (p3 ^ p1)
        + (i128)l3 * (p1 ^ p2);
    return det > 0;
}
```

8.6 Half Plane Intersection [c77d62]

```
// (int) needs polar angle
// arr must be nonempty; keeps the left of each line
auto area_pair(Line a, Line b) {
    return make_pair((a.b - a.a) ^ (b.a - a.a),
        (a.b - a.a) ^ (b.b - a.a));
}
bool isin(Line l0, Line l1, Line l2) {
    // Check inter(l1, l2) strictly in l0
    auto [a02X, a02Y] = area_pair(l0, l2);
    auto [a12X, a12Y] = area_pair(l1, l2);
    if (a12X < a12Y) a12X *= -1, a12Y *= -1;
    return (i128)a02Y * a12X > (i128)a02X * a12Y;
}
/* [solution exists] <=> [result.size() > 2] */
/* --^-- Line.a --^-- Line.b --^-- */
vector<Line> HalfPlaneInter(vector<Line> arr) {
    sort(arr.begin(), arr.end(), [&](Line a, Line b) {
        PT p1 = a.b - a.a, p2 = b.b - b.a;
        if (polar(p1, p2) != 0) return polar(p1, p2) < 0;
        return ori(a.a, a.b, b.b) < 0;
    });
    deque<Line> dq(1, arr[0]);
    auto pop_back = [&](int t, Line p) {
        while (sz(dq) >= t &&
            !isin(p, dq[sz(dq) - 2], dq.back()))
            dq.pop_back();
    };
    auto pop_front = [&](int t, Line p) {
        while (sz(dq) >= t && !isin(p, dq[0], dq[1]))
            dq.pop_front();
    };
    for (auto p : arr)
        if (polar(dq.back().b - dq.back().a,
            p.b - p.a) != 0)
            pop_back(2, p), pop_front(2, p),
            dq.push_back(p);
    pop_back(3, dq[0]), pop_front(3, dq.back());
    return vector<Line>(dq.begin(), dq.end());
}
```

8.7 Minkowski Sum [f559ee]

```
// (int) needs convex hull
// each input needs at least two distinct points
vector<PT> Minkowski(vector<PT> a, vector<PT> b) {
    a = ConvexHull(a), b = ConvexHull(b);
    int n = sz(a), m = sz(b);
    vector<PT> c{a[0] + b[0]}, s1, s2;
    For(i, 0, n - 1)
        s1.push_back(a[(i + 1) % n] - a[i]);
    For(i, 0, m - 1)
        s2.push_back(b[(i + 1) % m] - b[i]);
    for (int p1 = 0, p2 = 0; p1 < n || p2 < m;)
        if (p2 == m || (p1 < n &&
            sign(s1[p1] ^ s2[p2]) >= 0))
            c.push_back(c.back() + s1[p1++]);
        else c.push_back(c.back() + s2[p2++]);
    return ConvexHull(c);
}
```

8.8 Polar Angle [423cf8]

```
// (int) CCW from (1, 0), without length tie-breaking
int halfplane(PT p) {
    if (sign(p * p) == 0) return 0;
    bool upper = sign(p.y) > 0 ||
        (sign(p.y) == 0 && sign(p.x) > 0);
    return 1 - 2 * upper;
}
// upper(-1) -> origin(0) -> Lower(1)
strong_ordering polar(PT a, PT b) {
```

```

int ha = halfplane(a), hb = halfplane(b);
if (ha != hb) return ha <= hb;
return 0 <= sign(a ^ b);
}

```

8.9 Rotating Sweep Line [f1c4b0]

```

// (int) needs polar angle
// pts: 0-indexed Pt array
void RotSweepLine(int n, PT* pts) {
    using E = pair<PT, pii>;
    vector<E> ev; // dir, i, j: (i, j) => (j, i)
    For(i, 0, n - 1) For(j, 0, i - 1) {
        PT dir = pts[j] - pts[i];
        halfplane(dir) < 0 ? ev.push_back({dir, {i, j}})
        : ev.push_back({PT(0, 0) - dir, {j, i}});
    }
    sort(ev.begin(), ev.end(), [&](E e1, E e2) {
        auto pol = polar(e1.F, e2.F);
        return pol < 0 || (pol == 0 &&
            pll(e1.F * pts[e1.S.F], e1.F * pts[e1.S.S])
            < pll(e1.F * pts[e2.S.F], e1.F * pts[e2.S.S]));
    });
    vector<int> ord(n), rk(n);
    iota(ord.begin(), ord.end(), 0);
    sort(ord.begin(), ord.end(), [&](int i, int j) {
        return make_pair(pts[i].y, pts[i].x)
            < make_pair(pts[j].y, pts[j].x);
    });
    For(i, 0, n - 1) rk[ord[i]] = i;
    // init order with point indices ord[*]
    int ne = (int)ev.size();
    For(ie, 0, ne - 1) {
        int i, j; tie(i, j) = ev[ie].S;
        // update swap i, j
        rk[i]++; rk[j]--; // rk[point idx]: rank in ord
        tie(ord[rk[i]], ord[rk[i] - 1]) = tie(i, j);
        if (ie == ne - 1 ||
            polar(ev[ie + 1].F, ev[ie].F) != 0) {
            // update answer
        }
    }
}

```

8.10 Segment Intersect [118283]

```

bool HasInter(PT p1, PT p2, PT p3, PT p4) { // 4d829f
    int a123 = ori(p1, p2, p3), a124 = ori(p1, p2, p4);
    int a341 = ori(p3, p4, p1), a342 = ori(p3, p4, p2);
    if (!a123 && !a124)
        return btw(p1, p2, p3) || btw(p1, p2, p4) ||
            btw(p3, p4, p1) || btw(p3, p4, p2);
    return a123 * a124 <= 0 && a341 * a342 <= 0;
} // (int) does p1p2 intersect p3p4
PT GetInter(PT p1, PT p2, PT p3, PT p4) { // 2629b1
    double a123 = (p2 - p1) ^ (p3 - p1);
    double a124 = (p2 - p1) ^ (p4 - p1);
    return (p4 * a123 - p3 * a124) / (a123 - a124);
} // (FP) C^3 / C^2; requires nonparallel inputs

```

8.11 Circle Intersect With Any [c31e48]

```

// (FP) nondegenerate Lines; polygon edges != 0
vector<PT> CircleLineInter(Cir c, Line l) { // ab01ed
    PT p = l.a + (l.b - l.a) *
        ((c.o - l.a) * (l.b - l.a)) /
        abs2(l.b - l.a);
    double s = (l.b - l.a) ^ (c.o - l.a);
    double h2 = c.r * c.r - s * s /
        abs2(l.b - l.a);
    if (sign(h2) == -1) return {};
    if (sign(h2) == 0) return {p};
    PT h = (l.b - l.a) / abs(l.b - l.a) * sqrt(h2);
    return {p - h, p + h};
}
vector<PT> CirclesInter(Cir c1, Cir c2) { // e9fe3a
    double d2 = abs2(c1.o - c2.o), d = sqrt(d2);
    if (sign(d2) == 0) return {}; // concentric circles
    if (d < max(c1.r, c2.r) - min(c1.r, c2.r) ||
        d > c1.r + c2.r) return {};
    PT u = (c1.o + c2.o) / 2 + (c1.o - c2.o) *
        ((c2.r * c2.r - c1.r * c1.r) / (2 * d2));

```

```

    double A = sqrt((c1.r + c2.r + d) *
        (c1.r - c2.r + d) * (c1.r + c2.r - d) *
        (-c1.r + c2.r + d));
    PT v = PT{c1.o.y - c2.o.y, -c1.o.x + c2.o.x}
        * A / (2 * d2);
    if (sign(v.x) == 0 && sign(v.y) == 0) return {u};
    return {u + v, u - v};
}
double _area(PT pa, PT pb, double r) { // 1238aa
    if (abs(pa) < abs(pb)) swap(pa, pb);
    if (abs(pb) < EPS) return 0;
    double S, h, theta;
    double a = abs(pb), b = abs(pa), c = abs(pb - pa);
    double cosB = pb * (pb - pa) / a / c, B = acos(cosB);
    double cosC = (pa * pb) / a / b, C = acos(cosC);
    if (a > r) {
        S = (C / 2) * r * r;
        h = a * b * sin(C) / c;
        if (h < r && B < numbers::pi / 2)
            S -= acos(h / r) * r * r -
                h * sqrt(r * r - h * h);
    } else if (b > r) {
        theta = numbers::pi - B - asin(sin(B) / r * a);
        S = .5 * a * r * sin(theta) +
            (C - theta) / 2 * r * r;
    } else S = .5 * sin(C) * a * b;
    return S;
}
double AreaPolyCircle(vector<PT> poly, PT O,
    double r) { // 934b82
    double S = 0; int n = sz(poly);
    for (int i = 0; i < n; ++i)
        S += _area(poly[i] - O, poly[(i + 1) % n] - O,
            r) * ori(O, poly[i], poly[(i + 1) % n]);
    return fabs(S);
}

```

8.12 Tangents [b087af]

```

// (FP) circles have positive radii
PT unit(PT p) { return p / abs(p); }
PT ccw(PT p, double t) {
    return PT(
        p.x * cos(t) - p.y * sin(t),
        p.x * sin(t) + p.y * cos(t)
    );
}
vector<Line> Tangent(Cir c, PT p) { // bb4a23
    vector<Line> z;
    double d = abs(p - c.o);
    if (sign(d - c.r) == 0) {
        z.push_back({p, p + ccw90(p - c.o)});
    } else if (d > c.r) {
        double o = acos(c.r / d);
        PT i = unit(p - c.o);
        PT j = ccw(i, o) * c.r;
        PT k = ccw(i, -o) * c.r;
        z.push_back({c.o + j, p});
        z.push_back({c.o + k, p});
    }
    return z;
}
vector<Line> Tangent(Cir c1, Cir c2, int sign1) { //
    sign1 = 1 for outer, -1 for inter // d97c9a
    vector<Line> ret;
    double d_sq = abs2(c1.o - c2.o);
    if (sign(d_sq) == 0) return ret;
    double d = sqrt(d_sq);
    PT v = (c2.o - c1.o) / d;
    double c = (c1.r - sign1 * c2.r) / d;
    if (c * c > 1) return ret;
    double h = sqrt(max(0.0, 1.0 - c * c));
    for (int sign2 = 1; sign2 >= -1; sign2 -= 2) {
        PT n = PT{v.x * c - sign2 * h * v.y,
            v.y * c + sign2 * h * v.x};
        PT p1 = c1.o + n * c1.r;
        PT p2 = c2.o + n * (c2.r * sign1);
        if (sign(p1.x - p2.x) == 0 &&
            sign(p1.y - p2.y) == 0)
            p2 = p1 + ccw90(c2.o - c1.o);
        ret.push_back({p1, p2});
    }
}

```

```

return ret;
}

```

8.13 Tangent to Convex Hull [f1fdc2]

```

// (int) needs cyclic ternary search
// C is nonempty; p is strictly outside C
pii HullTangent(vector<PT> &C, PT p) {
    auto gao = [&](int s) {
        return cyc_tsearch(sz(C), [&](int x, int y)
            { return ori(p, C[x], C[y]) == s; });
    };
    return pii(gao(1), gao(-1));
} // return (a, b), ori(p, C[a], C[b]) >= 0

```

8.14 Minimum Enclosing Circle [07df39]

```

// (FP) needs segment intersect; p must be nonempty
PT circenter(PT a, PT b, PT c) {
    PT ab = (a + b) / 2, ac = (a + c) / 2;
    return GetInter(
        ab, ab + ccw90(b - a),
        ac, ac + ccw90(c - a)
    );
}
Cir MinEnclosing(vector<PT> &p) {
    shuffle(iter(p), mt19937(clock()));
    double r = 0;
    PT cent = p[0];
    For(i, 0, sz(p) - 1) if (abs2(cent - p[i]) > r) {
        cent = p[i]; r = 0;
        For(j, 0, i - 1) if (abs2(cent - p[j]) > r) {
            cent = (p[i] + p[j]) / 2;
            r = abs2(p[j] - cent);
            For(k, 0, j - 1) if (abs2(cent - p[k]) > r) {
                cent = circenter(p[i], p[j], p[k]);
                r = abs2(p[k] - cent);
            }
        }
    }
    return {cent, sqrt(r)};
}

```

8.15 Union of Stuff [ae9b12]

```

// (FP) needs segment intersect
// circles are pairwise distinct; polygons are C CW
vector<pair<double, double>> CoverSegment(Cir a, Cir b)
{ // 7aelbe
    double d = abs(a.o - b.o);
    vector<pair<double, double>> res;
    if (sign(a.r + b.r - d) == 0);
    else if (d <= abs(a.r - b.r) + EPS) {
        if (a.r < b.r)
            res.emplace_back(0, 2 * numbers::pi);
    } else if (d < abs(a.r + b.r) - EPS) {
        double o = acos((a.r * a.r + d * d - b.r * b.r)
            / (2 * a.r * d));
        double z = atan2((b.o - a.o).y, (b.o - a.o).x);
        if (z < 0) z += 2 * numbers::pi;
        double l = z - o, r = z + o;
        if (l < 0) l += 2 * numbers::pi;
        if (r > 2 * numbers::pi) r -= 2 * numbers::pi;
        if (l > r)
            res.emplace_back(l, 2 * numbers::pi),
            res.emplace_back(0, r);
        else res.emplace_back(l, r);
    }
    return res;
}
double CircleUnionArea(vector<Cir> c) { // e653e9
    int n = sz(c);
    double a = 0, w;
    for (int i = 0; w = 0, i < n; ++i) {
        vector<pair<double, double>> s = {
            {2 * numbers::pi, 9}
        }, z;
        for (int j = 0; j < n; ++j) if (i != j) {
            z = CoverSegment(c[i], c[j]);
            for (auto &e : z) s.push_back(e);
        }
        sort(s.begin(), s.end());
    }
}

```

```

auto F = [&] (double t) {
    return c[i].r * (c[i].r * t +
        c[i].o.x * sin(t) - c[i].o.y * cos(t));
};
for (auto &e : s) {
    if (e.first > w) a += F(e.first) - F(w);
    w = max(w, e.second);
}
return a * 0.5;
}
// Union of Polygons
double PolyUnion(vector<vector<PT>> poly) { // 827711
    int n = sz(poly);
    double ans = 0;
    auto solve = [&](PT a, PT b, int cid) {
        vector<pair<PT, int>> event;
        for (int i = 0; i < n; ++i) {
            int st = 0, m = sz(poly[i]);
            while (st < m && ori(poly[i][st], a, b) != 1)
                st++;
            if (st == m) continue;
            For(j, 0, m - 1) {
                PT c = poly[i][(j + st) % m];
                PT d = poly[i][(j + st + 1) % m];
                if (sign((a - b) ^ (c - d)) != 0) {
                    int ok1 = ori(c, a, b) == 1;
                    int ok2 = ori(d, a, b) == 1;
                    if (ok1 ^ ok2)
                        event.emplace_back(GetInter(a, b, c, d),
                            ok1 ? 1 : -1);
                } else if (ori(c, a, b) == 0 &&
                    sign((a - b) * (c - d)) > 0 && i <= cid) {
                    event.emplace_back(c, -1);
                    event.emplace_back(d, 1);
                }
            }
        }
        sort(iter(event), [&](pair<PT, int> i,
            pair<PT, int> j) {
            return (a - i.first) * (a - b) <
                (a - j.first) * (a - b);
        });
        int now = 0;
        PT lst = a;
        for (auto [x, y] : event) {
            if (btw(a, b, lst) && btw(a, b, x) && !now)
                ans += lst ^ x;
            now += y, lst = x;
        }
        For(i, 0, n - 1) For(j, 0, sz(poly[i]) - 1)
            solve(poly[i][j], poly[i][(j + 1) % sz(poly[i])],
                i);
        return ans / 2;
    }
}

```

8.16 Delaunay Triangulation [1c018f]

```

// (int) needs point in circle, segment intersect
const int N = 0; // set per problem constraints
// needs at least one distinct point
/* Delaunay Triangulation:
Given a sets of points on 2D plane, find a
triangulation such that no points will strictly
inside circumcircle of any triangle. */
struct Edge {
    int id; // oidx[id]
    list<Edge>::iterator twin;
    Edge(int _id = 0):id(_id) {}
};
struct Delaunay { // 0-base
    int n, oidx[N];
    list<Edge> head[N]; // result udir. graph
    PT p[N];
    void init(int _n, PT _p[]) {
        n = _n, iota(oidx, oidx + n, 0);
        for (int i = 0; i < n; ++i) head[i].clear();
        sort(oidx, oidx + n, [&](int a, int b) {
            return _p[a] < _p[b];
        });
        for (int i = 0; i < n; ++i) p[i] = _p[oidx[i]];
    }
}

```

```

    divide(0, n - 1);
}
void addEdge(int u, int v) {
    head[u].push_front(Edge(v));
    head[v].push_front(Edge(u));
    head[u].begin()->twin = head[v].begin();
    head[v].begin()->twin = head[u].begin();
}
void divide(int l, int r) {
    if (l == r) return;
    if (l + 1 == r) return addEdge(l, l + 1);
    int mid = (l + r) >> 1, nw[2] = {l, r};
    divide(l, mid), divide(mid + 1, r);
    auto gao = [&](int t) {
        PT pt[2] = {p[nw[0]], p[nw[1]]};
        for (auto it : head[nw[t]]) {
            int v = ori(pt[1], pt[0], p[it.id]);
            if (v > 0 || (v == 0 &&
                (pt[t ^ 1] - p[it.id]) *
                (pt[t ^ 1] - p[it.id]) <
                (pt[1] - pt[0]) * (pt[1] - pt[0])))
                return nw[t] = it.id, true;
        }
        return false;
    };
    while (gao(0) || gao(1));
    addEdge(nw[0], nw[1]); // add tangent
    while (true) {
        PT pt[2] = {p[nw[0]], p[nw[1]]};
        int ch = -1, sd = 0;
        for (int t = 0; t < 2; ++t)
            for (auto it : head[nw[t]])
                if (ori(pt[0], pt[1], p[it.id]) > 0 &&
                    (ch == -1 || InCir(pt[0], pt[1],
                        p[ch], p[it.id])))
                    ch = it.id, sd = t;
        if (ch == -1) break; // upper common tangent
        for (auto it = head[nw[sd]].begin();
            it != head[nw[sd]].end(); )
            if (HasStrictInter(pt[sd], p[it->id],
                pt[sd ^ 1], p[ch]))
                head[it->id].erase(it->twin),
                head[nw[sd]].erase(it++);
            else ++it;
        nw[sd] = ch, addEdge(nw[0], nw[1]);
    }
}
} tool;

```

8.17 Voronoi Diagram [c6bcb2]

```

// (int) needs delaunay triangulation
// all coord is even, may call HalfPlaneInter after
vector<vector<Line>> vec;
void BuildVoronoiLine(int n, PT *arr) {
    tool.init(n, arr); // Delaunay
    vec.clear(), vec.resize(n);
    for (int i = 0; i < n; ++i)
        for (auto e : tool.head[i]) {
            int u = tool.oidx[i], v = tool.oidx[e.id];
            PT m((arr[v].x + arr[u].x) / 2,
                (arr[v].y + arr[u].y) / 2);
            PT d = ccw90(arr[v] - arr[u]);
            vec[u].push_back({m, m + d});
        }
}

```

8.18 3D Basic [4a7a37]

```

// (FP) add Point*scalar for proj
struct Point { double x = 0, y = 0, z = 0; };
Point operator-(Point p1, Point p2)
{return Point{p1.x-p2.x, p1.y-p2.y, p1.z-p2.z};}
Point operator+(Point p1, Point p2)
{return Point{p1.x+p2.x, p1.y+p2.y, p1.z+p2.z};}
Point operator/(Point p1, double v)
{return Point{p1.x / v, p1.y / v, p1.z / v};}
Point operator^(Point p1, Point p2) {
    return Point{p1.y * p2.z - p1.z * p2.y,
        p1.z * p2.x - p1.x * p2.z,
        p1.x * p2.y - p1.y * p2.x};
}

```

```

double operator*(Point p1, Point p2)
{ return p1.x * p2.x + p1.y * p2.y + p1.z * p2.z; }
double abs(Point a) { return sqrt(a * a); }
Point cross3(Point a, Point b, Point c)
{ return (b - a) ^ (c - a); }
double area2(Point a, Point b, Point c)
{ return abs(cross3(a, b, c)); }
double volume(Point a, Point b, Point c, Point d)
{ return cross3(a, b, c) * (d - a); }
Point Masscenter(Point a, Point b, Point c, Point d)
{ return (a + b + c + d) / 4; }
PT proj(Point a, Point b, Point c, Point u) {
    // proj. u to the plane of a, b, and c
    Point e1 = b - a;
    Point e2 = c - a;
    e1 = e1 / abs(e1);
    e2 = e2 - e1 * (e2 * e1);
    e2 = e2 / abs(e2);
    Point p = u - a;
    return PT{p * e1, p * e2};
}

```

8.19 3D Convex Hull [166108]

```

// (FP) needs 3D basic
// add Point==Point; returns {} unless hull is 3D
struct Face {
    int a, b, c;
    Face(int _a, int _b, int _c) : a(_a), b(_b), c(_c) {}
};
auto preprocess(auto pts) {
    auto G = pts.begin();
    vector<int> id;
    auto fail = tuple{-1, -1, -1, id};
    int a = int(find_if(iter(pts), [&](Point z) {
        return z != *G;
    }) - G);
    if (a == sz(pts)) return fail;
    int b = int(find_if(iter(pts), [&](Point z) {
        return cross3(*G, pts[a], z) != Point{0, 0, 0};
    }) - G);
    if (b == sz(pts)) return fail;
    int c = int(find_if(iter(pts), [&](Point z) {
        return sign(volume(*G, pts[a], pts[b], z)) != 0;
    }) - G);
    if (c == sz(pts)) return fail;
    for (int i = 0; i < sz(pts); i++)
        if (i != a && i != b && i != c)
            id.push_back(i);
    return tuple{a, b, c, id};
}
// return the faces with pts indexes
vector<Face> ConvexHull3D(vector<Point> pts) {
    int n = sz(pts);
    if (n <= 3) return {}; // be careful about edge case
    vector<Face> now;
    vector<vector<int>> z(n, vector<int>(n));
    auto [a, b, c, ord] = preprocess(pts);
    if (a == -1) return {};
    now.emplace_back(a, b, c); now.emplace_back(c, b, a);
    for (auto i : ord) {
        vector<Face> nxt;
        for (auto &f : now) {
            auto v = volume(pts[f.a], pts[f.b],
                pts[f.c], pts[i]);
            if (sign(v) <= 0) nxt.push_back(f);
            z[f.a][f.b] = z[f.b][f.c] =
                z[f.c][f.a] = sign(v);
        }
        auto F = [&](int x, int y) {
            if (z[x][y] > 0 && z[y][x] <= 0)
                nxt.emplace_back(x, y, i);
        };
        for (auto &f : now)
            F(f.a, f.b), F(f.b, f.c), F(f.c, f.a);
        now = nxt;
    }
    return now;
}
// n^2 delaunay: facets with negative z normal of
// convexhull of (x, y, x^2 + y^2), use a pseudo-point
// (0, 0, inf) to avoid degenerate case

```

```
// test @ SPOJ CH3D
// double area = 0, vol = 0; // surface area / volume
// for (auto [a, b, c]: faces)
//     area += abs(ver(p[a], p[b], p[c]))/2.0,
//     vol += volume(P3(0, 0, 0), p[a], p[b], p[c])/6.0;
```

9 Misc

9.1 All LCS [ba9cc9]

```
void all_lcs(string s, string t) { // 0-base
    vector<int> h(sz(t));
    iota(all(h), 0);
    for (int a = 0; a < sz(s); ++a) {
        int v = -1;
        for (int c = 0; c < sz(t); ++c)
            if (s[a] == t[c] || h[c] < v)
                swap(h[c], v);
        // LCS(s[0, a], t[b, c]) =
        // c - b + 1 - sum([h[i] >= b] | i <= c)
        // h[i] might become -1 !!
    }
}
```

9.2 Binary Search On Fraction [765c5a]

```
struct Q {
    ll p, q;
    Q go(Q b, ll d) { return {p + b.p*d, q + b.q*d}; }
};
bool pred(Q);
// returns smallest p/q in [lo, hi] such that
// pred(p/q) is true, and 0 <= p,q <= N
Q frac_bs(ll N) {
    Q lo{0, 1}, hi{1, 0};
    if (pred(lo)) return lo;
    assert(pred(hi));
    bool dir = 1, L = 1, H = 1;
    for (; L || H; dir = !dir) {
        ll len = 0, step = 1;
        for (int t = 0; t < 2 && (t ? step/=2 : step*=2);)
            if (Q mid = hi.go(lo, len + step);
                mid.p > N || mid.q > N || dir ^ pred(mid))
                t++;
            else len += step;
        swap(lo, hi = hi.go(lo, len));
        (dir ? L : H) = !!len;
    }
    return dir ? hi : lo;
}
```

9.3 Cyclic Ternary Search [9017cc]

```
/* bool pred(int a, int b);
f(0) ~ f(n-1) is a cyclic-shift U-function
return idx s.t. pred(x, idx) is false forall x*/
int cyc_tsearch(int n, auto pred) {
    if (n == 1) return 0;
    int l = 0, r = n; bool rv = pred(1, 0);
    while (r - l > 1) {
        int m = (l + r) / 2;
        if (pred(0, m) ? rv : pred(m, (m + 1) % n)) r = m;
        else l = m;
    }
    return pred(1, r % n) ? l : r % n;
}
```

9.4 Matroid Intersection

$M = (E, \mathcal{I})$, where $\mathcal{I} \subseteq 2^E$ is nonempty, is a matroid if:

- If $S \in \mathcal{I}$ and $S' \subsetneq S$, then $S' \in \mathcal{I}$.
- For $S_1, S_2 \in \mathcal{I}$ s.t. $|S_1| < |S_2|$, there exists $e \in S_2 \setminus S_1$ s.t. $S_1 \cup \{e\} \in \mathcal{I}$.

Matroid intersection:

Start from $S = \emptyset$. In each iteration, let

- $Y_1 = \{x \notin S \mid S \cup \{x\} \in \mathcal{I}_1\}$
- $Y_2 = \{x \notin S \mid S \cup \{x\} \in \mathcal{I}_2\}$

If there exists $x \in Y_1 \cap Y_2$, insert x into S . Otherwise for each $x \in S, y \notin S$, create edges

- $x \rightarrow y$ if $S - \{x\} \cup \{y\} \in \mathcal{I}_1$.
- $y \rightarrow x$ if $S - \{x\} \cup \{y\} \in \mathcal{I}_2$.

Find a *shortest* path (with BFS) starting from a vertex in Y_1 and ending at a vertex in Y_2 which doesn't pass through any other vertices in Y_2 , and alternate the path. The size of S will be incremented by 1 in each iteration. For the weighted case, assign weight $w(x)$ to vertex x if $x \in S$ and $-w(x)$ if $x \notin S$. Find the path with the minimum number of edges among all minimum length paths and alternate it.

9.5 Min Plus Convolution [081e9d]

```
// a is convex a[i+1]-a[i] <= a[i+2]-a[i+1]
vector<int> min_plus_convolution(vector<int> &a, vector
<int> &b) {
    int n = sz(a), m = sz(b);
    vector<int> c(n + m - 1, INF);
    auto dc = [&](auto Y, int l, int r, int jl, int jr) {
        if (l > r) return;
        int mid = (l + r) / 2, from = -1, &best = c[mid];
        for (int j = jl; j <= jr; ++j)
            if (int i = mid - j; i >= 0 && i < n)
                if (best > a[i] + b[j])
                    best = a[i] + b[j], from = j;
        Y(Y, l, mid - 1, jl, from), Y(Y, mid + 1, r, from, jr);
    };
    return dc(dc, 0, n - 1 + m - 1, 0, m - 1), c;
}
```

9.6 Mo's Algorithm [ea5261]

```
struct MoAlgorithm {
    struct query {
        int l, r, id;
        bool operator < (const query &o) {
            if (l / C == o.l / C)
                return (l / C) & 1 ? r > o.r : r < o.r;
            return l / C < o.l / C;
        }
    };
    int cur_ans;
    vector<int> ans;
    void add(int x) {} // do something
    void sub(int x) {} // do something
    vector<query> Q;
    void add_query(int l, int r, int id) { // [L, r)
        Q.push_back({l, r, id});
        ans.push_back(0);
    }
    void run() {
        sort(Q.begin(), Q.end());
        int pl = 0, pr = 0;
        cur_ans = 0;
        for (query &i : Q) {
            while (pl > i.l) add(a[--pl]);
            while (pr < i.r) add(a[pr++]);
            while (pl < i.l) sub(a[pl++]);
            while (pr > i.r) sub(a[--pr]);
            ans[i.id] = cur_ans;
        }
    }
};
```

9.7 Mo's Algorithm On Tree [8331c2]

```
/* Mo's Algorithm On Tree
Preprocess:
1) LCA
2) dfs with in[u] = dft++, out[u] = dft++
3) ord[in[u]] = ord[out[u]] = u
4) bitset<MAXN> inset */
struct Query {
    int L, R, Lbid, lca;
    Query(int u, int v) {
        int c = LCA(u, v);
        if (c == u || c == v)
            q.lca = -1, q.L = out[c ^ u ^ v], q.R = out[c];
        else if (out[u] < in[v])
            q.lca = c, q.L = out[u], q.R = in[v];
        else
            q.lca = c, q.L = out[v], q.R = in[u];
        q.Lid = q.L / blk;
    }
};
```

```

bool operator<(const Query &q) const {
    if (LBid != q.LBid) return LBid < q.LBid;
    return R < q.R;
}
};
void flip(int x) {
    if (inset[x]) sub(arr[x]); // TODO
    else add(arr[x]); // TODO
    inset[x] = ~inset[x];
}
void solve(vector<Query> query) {
    sort(ALL(query));
    int L = 0, R = 0;
    for (auto q : query) {
        while (R < q.R) flip(ord[++R]);
        while (L > q.L) flip(ord[--L]);
        while (R > q.R) flip(ord[R--]);
        while (L < q.L) flip(ord[L++]);
        if (~q.lca) add(arr[q.lca]);
        // answer query
        if (~q.lca) sub(arr[q.lca]);
    }
}

```

9.8 PBDS [d65996]

```

#include <ext/pb_ds/tree_policy.hpp>
#include <ext/pb_ds/assoc_container.hpp>
using namespace __gnu_pbds;
tree<int, null_type, less<int>, rb_tree_tag,
    tree_order_statistics_node_update> oset;
// order_of_key, find_by_order
cc_hash_table<int, int> m1;
gp_hash_table<int, int> m2;
// Like map, but much faster

```

9.9 Simulated Annealing [de78c6]

```

double factor = 100000;
const int base = 1e9; // remember to run ~ 10 times
for (int it = 1; it <= 1000000; ++it) {
    // ans: answer, nw: current value, rnd(): mt19937
    if (exp(-(nw - ans) / factor) >= (double)rnd() %
        base) / base) ans = nw;
    factor *= 0.99995;
}

```

9.10 SOS dp [6aadb1]

```

//memory optimized, super easy to code.
rep(i, (1 << N)) F[i] = A[i];
rep(i, N) rep(mask, (1 << N)) {
    if (mask & (1<<i)) F[mask] += F[mask^(1<<i)];
}

```

9.11 SMAWK [e3ace9]

```

// For all 2x2 submatrix:
// If M[1][0] < M[1][1], M[0][0] < M[0][1]
// If M[1][0] == M[1][1], M[0][0] <= M[0][1]
// M[i][ans_i] is the best value in the i-th row
// select(int r, int u, int v) return true if f(r, v)
// is better than f(r, u)
vector<int> smawk(int N, int M, auto &&select) {
    auto dc = [&](auto self, const vector<int> &r, const
        vector<int> &c) {
        if (r.empty()) return vector<int>{};
        const int n = sz(r); vector<int> ans(n), nr, nc;
        for (int i : c) {
            while (!nc.empty() &&
                select(r[nc.size() - 1], nc.back(), i))
                nc.pop_back();
            if (int(nc.size()) < n) nc.push_back(i);
        }
        for (int i = 1; i < n; i += 2) nr.push_back(r[i]);
        const auto na = self(self, nr, nc);
        for (int i = 1; i < n; i += 2) ans[i] = na[i >> 1];
        for (int i = 0, j = 0; i < n; i += 2) {
            ans[i] = nc[j];
            const int end = i + 1 == n ? nc.back() : ans[i +
                1];
            while (nc[j] != end)

```

```

        if (select(r[i], ans[i], nc[++j])) ans[i] = nc[
            j];
        }
        return ans;
    };
    vector<int> R(N), C(M); iota(iter(R), 0), iota(iter(C)
        ), 0);
    return dc(dc, R, C);
}

```

9.12 Tree Hash [34aae5]

```

ull seed;
ull shift(ull x) {
    x ^= x << 13;
    x ^= x >> 7;
    x ^= x << 17;
    return x;
}
ull dfs(int u, int f) {
    ull sum = seed;
    for (int i : G[u]) if (i != f)
        sum += shift(dfs(i, u));
    return sum;
}

```

9.13 Python [ebfb5e]

```

from [decimal, fractions, math, random] import *
setcontext(Context(prec=10, Emax=MAX_EMAX, rounding=
    ROUND_FLOOR))
Decimal('1.1') / Decimal('0.2')
Fraction(3, 7)
Fraction(Decimal('1.14'))
Fraction('1.2').limit_denominator(4).numerator
Fraction(cos(pi / 3)).limit_denominator()
print(*[randint(1, C) for i in range(0, N)], sep=' ')

```